

Human DCP1 is crucial for mRNA decapping and possesses paralog-specific gene regulating functions

Reviewed Preprint

v2 • October 7, 2024

Revised by authors

Reviewed Preprint

v1 • February 5, 2024

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Abstract

The mRNA 5'-cap structure removal by the decapping enzyme DCP2 is a critical step in gene regulation. While DCP2 is the catalytic subunit in the decapping complex, its activity is strongly enhanced by multiple factors, particularly DCP1, which is the major activator in yeast. However, the precise role of DCP1 in metazoans has yet to be fully elucidated. Moreover, in humans, the specific biological functions of the two DCP1 paralogs, DCP1a and DCP1b, remain largely unknown. To investigate the role of human DCP1, we generated cell-lines that were deficient in DCP1a, DCP1b or both to evaluate the importance of DCP1 in the decapping machinery. Our results highlight the importance of human DCP1 in decapping process and show that the EVH1 domain of DCP1 enhances the mRNA-binding affinity of DCP2. Transcriptome and metabolome analyses outline the distinct functions of DCP1a and DCP1b in human cells, regulating specific endogenous mRNA targets and biological processes. Overall, our findings provide insights into the molecular mechanism of human DCP1 in mRNA decapping and shed light on the distinct functions of its paralogs.

eLife Assessment

This study attempts to understand the functional roles of the human DCP1 paralogs in regulating RNA decay by DCP2. Using a combination of cellular-based assays and in vitro assays, the authors conclude that DCP1a/b plays a role in regulating DCP2 activity. While this revised version presents some new and interesting observations on human DCP1, the underlying data to support its claims remain **incomplete**. Overall, these results will be **useful** to the RNA community.

<https://doi.org/10.7554/eLife.94811.2.sa2>

Introduction

Gene expression must be stringently regulated to ensure efficient control of cellular processes. A key mechanism of gene expression regulation is the control of mRNA degradation (Chen and Shyu, 2017; Corbett, 2018), mediated by the removal of protective structures from mRNA, which suppresses translation or leads to mRNA degradation. In most eukaryotic cells, mRNA is stabilized through the addition during posttranscriptional modification of an N^7 -methyl guanosine (m^7G) cap at the 5' end and a polyA tail at the 3' end (Furuichi et al., 1977; Garneau et al., 2007). General mRNA decay is typically initiated by deadenylation, which is mediated by the coordinated activity of the CCR4/NOT and PAN2/3 complexes (Chen and Shyu, 2011; Decker and Parker, 1993). Shortening of the polyA tail typically initiates mRNA decay, resulting in exonucleolytic degradation in the 5'-3' or 3'-5' direction, respectively facilitated by the exonuclease XRN1 or the exosome complex (Chang et al., 2019; Garneau et al., 2007; Lim et al., 2016; Subtelny et al., 2014; Webster et al., 2018; Yi et al., 2018). While the cytoplasmic RNA exosome degrades mRNA in the 3'-5' direction, the primary effect of deadenylation is “mRNA decapping”, the removal of the m^7G cap structure at the 5'-end of the mRNA that otherwise prevents mRNA degradation (Stevens, 1988). This connection between deadenylation and decapping is facilitated by Pat1, which together with the Lsm1-7 complex, recruits the decapping complex through a combination of disordered and structured domains, promoting distinct steps of mRNA decay (Charenton et al., 2017; Charenton and Graille, 2018; Fourati et al., 2014; Lobel and Gross, 2020; Lobel et al., 2019; Sharif and Conti, 2013; Wu et al., 2014).

The mRNA decapping process is crucial for gene regulation because it leads to the irreversible removal of the mRNA cap structure, inhibiting translation initiation and inducing the XRN1-mediated degradation of mRNA through the 5'-3' mRNA decay pathway (Hsu and Stevens, 1993; Topisirovic et al., 2011). Decapping does not only lead to the degradation of bulk mRNA molecules; it is also involved in specific mRNA decay pathways, such as AU-rich element-mediated decay, nonsense-mediated decay (NMD), and miRNA-triggered mRNA turnover (Arribas-Layton et al., 2013; Chen and Shyu, 2003; Fenger-Gron et al., 2005; Jonas and Izaurralde, 2015; Loh et al., 2013). Because of its crucial role in gene regulation, decapping is tightly regulated by the activity of both general and pathway-specific factors through various mechanisms. In the context of NMD for instance, target transcripts can be degraded through endonucleolytic cleavage by SMG6 (Eberle et al., 2009; Huntzinger et al., 2008), or be facilitated by SMG5 in conjunction with the NMD effector SMG7 or the decapping factor PNR2, leading to enhanced deadenylation and decapping (Cho et al., 2009; Fukuhara et al., 2005; Loh et al., 2013). Understanding the determinants of the preferential associations between certain degradation mechanisms and particular transcripts requires further study (Metze et al., 2013). Overall, the pivotal role of decapping in the intricate network of mRNA decay justifies ongoing research in this field.

In eukaryotes, the removal of the mRNA 5'- m^7GDP cap structure is catalysed by the decapping enzyme DCP2, a pyrophosphatase belonging to the Nudix family of proteins. This process results in the generation of 5' monophosphorylated mRNA, which is subsequently targeted for degradation (Arribas-Layton et al., 2013; Lykke-Andersen, 2002; Steiger et al., 2003; van Dijk et al., 2002; Wang et al., 2002). The central component of the complex, DCP2, has two major domains: the N-terminal regulatory domain (NRD) and the catalytic Nudix domain (She et al., 2006; She et al., 2008). The positively charged patch of the Nudix domain plays a crucial role in RNA binding (Deshmukh et al., 2008; Piccirillo et al., 2003). Meanwhile, the NRD enhances decapping activity by specifically recognizing the m^7G nucleotide of the mRNA cap structure and interacts with the N-terminal EVH1 domain of the major activator DCP1, further accelerating the decapping process (Chang et al., 2014; Charenton et al., 2016; Dunkley and Parker, 1999; Floor et al., 2010; Mugridge et al., 2016; She et al., 2008; Wang et al., 2002; Wurm et al., 2017). Yeast Dcp2 has an additional unstructured C-terminal extension that contains short

helical leucine-rich motifs (HLMs) responsible for protein-protein interactions and elements that inhibit Dcp2 catalytic activity (Charenton *et al.*, 2017 [↗](#); He and Jacobson, 2015 [↗](#)). Interestingly, in contrast to yeast, metazoans have the HLM in DCP1, implying a conserved biological function with evolving regulation of mRNA decapping (Fromm *et al.*, 2012 [↗](#)).

The interactions between DCP1 and DCP2 proteins and their functions across various species have garnered extensive attention in the study of mRNA degradation mechanisms. While DCP2 is capable of demonstrating catalytic activity on its own (Lykke-Andersen, 2002 [↗](#); van Dijk *et al.*, 2002 [↗](#); Wang *et al.*, 2002 [↗](#)), it is important to note that additional cofactors are involved in stimulating DCP2 for complete activation. In yeast, the primary activator of DCP2 is Dcp1, which is essential for decapping activity *in vivo* (Beelman *et al.*, 1996 [↗](#)) and considerably enhances the catalytic activity of Dcp2 *in vitro* (Deshmukh *et al.*, 2008 [↗](#); Floor *et al.*, 2010 [↗](#); She *et al.*, 2006 [↗](#)). Moreover, DCP1 has been shown to interact with other decapping proteins, such as DCP2, to form functional decapping complexes critical for mRNA degradation. Localization and functional analysis of decapping proteins across various species indicate that Dcp1-dependent decapping mechanisms have been conserved throughout evolution and play a significant role in the mRNA decapping machinery (Lall *et al.*, 2005 [↗](#); Sakuno *et al.*, 2004 [↗](#)). Furthermore, the importance of DCP1 in plant species underscores the necessity of the decapping complex for proper plant development and highlights its potential homology with yeast and human DCP1 (Xu *et al.*, 2006 [↗](#)). These findings suggest that the interaction between DCP1 and DCP2 and their role in mRNA degradation is a conserved key biological process across species.

In addition to Dcp1, proteins such as Edc1-3, Pby1, Lsm1-7, Pat1, and Dhh1 (PNRC1, PNRC2, EDC3, EDC4, Lsm1-7, Lsm14, PatL1, and DDX6 in metazoans) promote decapping by directly or indirectly interacting with Dcp2 (Charenton *et al.*, 2020 [↗](#); Charenton *et al.*, 2017 [↗](#); Charenton and Graille, 2018 [↗](#); Charenton *et al.*, 2016 [↗](#); Cho *et al.*, 2009 [↗](#); Gaviraghi *et al.*, 2018 [↗](#); He and Jacobson, 2022 [↗](#); He *et al.*, 2022 [↗](#); Jonas and Izaurralde, 2013 [↗](#); Ling *et al.*, 2011 [↗](#); Lobel *et al.*, 2019 [↗](#); Mugridge *et al.*, 2018b [↗](#)). These proteins facilitate decapping through the assembly of messenger ribonucleoprotein (RNP) units or the structural rearrangement of DCP2. High levels of these proteins have been detected in large RNP granules, processing bodies (P-bodies), which likely serve as sites for the storage of translationally repressed RNAs and the process of mRNA decay (Eulalio *et al.*, 2007 [↗](#); Jonas and Izaurralde, 2013 [↗](#); Luo *et al.*, 2018 [↗](#)).

The interactions between the subunits of the decapping complex are crucial for enhancing its activity and for the targeting of specific mRNAs. Although Dcp1 and Dcp2 have been shown to interact strongly in yeast, direct DCP1-DCP2 interactions are weak in metazoans. Instead, the metazoan-specific protein EDC4, which has binding sites for both DCP1 and DCP2, likely functions as a scaffold that brings these proteins together despite their weak interaction propensity (Chang *et al.*, 2014 [↗](#); Fenger-Gron *et al.*, 2005 [↗](#)). In addition to EDC4, DCP1 acts as a binding platform for other coactivator proteins, such as yeast Edc1 and Edc2 and the human PNRC paralogs, PNRC1 and PNRC2 (Gaviraghi *et al.*, 2018 [↗](#); Jonas and Izaurralde, 2013 [↗](#); Mugridge *et al.*, 2016 [↗](#)). In metazoans, the HLM domain of DCP1 interacts with the Lsm domains of proteins such as EDC3 and Lsm14 in a mutually exclusive manner. Moreover, in multicellular eukaryotes, DCP1 features a C-terminal extension that induces DCP1 trimerization, referred to as the trimerization domain (TD). Trimerization is a prerequisite for the incorporation of DCP1 into active decapping complexes and for promoting efficient mRNA decapping *in vivo* (Fromm *et al.*, 2012 [↗](#); Jonas and Izaurralde, 2013 [↗](#); Tritschler *et al.*, 2007 [↗](#)). This competition results in the formation of structurally and functionally distinct assemblies, contributing to target specificity (Vidya and Duchaine, 2022 [↗](#)).

Recent structural analyses of the *Kluyveromyces lactis* Dcp1-Dcp2-Edc3 complex indicate that the Lsm domain of Edc3 binds to an extended helix at the C-terminus of the Dcp2 Nudix domain (Charenton *et al.*, 2016 [↗](#)). Furthermore, the activation peptide of Edc1 bridges Dcp1 and Dcp2, as observed in the Edc1-Dcp1-Dcp2 complex in *Saccharomyces pombe* and *Kluyveromyces lactis* (Mugridge *et al.*, 2018b [↗](#); Wurm *et al.*, 2017 [↗](#)). These structural insights suggest that Edc1 and

Edc3 collaborate through distinct mechanisms, aiding the decapping process (Charenton and Graille, 2018 [DOI](#); Charenton *et al.*, 2016 [DOI](#); Mugridge *et al.*, 2018a [DOI](#); Mugridge *et al.*, 2018b [DOI](#); Tibble *et al.*, 2021 [DOI](#)). Specifically, Edc1 stabilizes Dcp2's catalytically active conformation, while Edc3 extends the RNA binding surface of the Dcp2 Nudix domain (Charenton *et al.*, 2016 [DOI](#); Mugridge *et al.*, 2018b [DOI](#)). This extended RNA interaction potentially accounts for the transcript-specific nature of Edc3-mediated decapping activation, hinting at distinct coactivators' roles in gene-specific regulation.

While structural studies suggest that interactions with Dcp1 are central to the activation of Dcp2 (Charenton and Graille, 2018 [DOI](#); Deshmukh *et al.*, 2008 [DOI](#); Mugridge *et al.*, 2018a [DOI](#); Mugridge *et al.*, 2016 [DOI](#); She *et al.*, 2008 [DOI](#); Valkov *et al.*, 2016 [DOI](#)), the lack of structural information on the human decapping complex hinders our understanding of the specific role of human DCP1 in decapping. It remains unclear how DCP1 contributes to the conformational changes in the DCP2 catalytic cycle or to mRNA and cap structure recognition. A further complication is the existence of two paralogs of human DCP1, DCP1a and DCP1b, which have 31% sequence identity over their entire length (Lykke-Andersen, 2002 [DOI](#); van Dijk *et al.*, 2002 [DOI](#)). The functional differences of these paralogs have yet to be determined. Studies of their potentially distinct roles in the decapping complex may provide new insight into the molecular mechanisms of mRNA decapping. The fact that decapping dysfunctions can have detrimental effects on cell growth and have been linked to severe neurological disorders in humans (Ahmed *et al.*, 2015 [DOI](#); Ng *et al.*, 2015 [DOI](#)), highlights the importance of understanding the mechanisms of mRNA decapping regulation and the potential therapeutic implications.

In this study, we investigated the roles of human DCP1 in mRNA decapping by generating HEK-293T cell lines lacking DCP1a, DCP1b, or both. Our findings indicate that human DCP1 promotes decapping *in vivo* and that DCP1a and DCP1b play redundant roles in the general 5'-3' mRNA decay pathway. We found that EVH1 domain may be crucial for enhancing DCP2's recruitment to target mRNA in cells. Multiomics analyses revealed distinct expressional and metabolomic profiles for the knockout cell lines. These results indicate that the DCP1 paralogs have distinct roles in the regulation of various biological processes. Overall, our findings offer valuable insights into the mechanisms underlying human DCP1-mediated mRNA decapping and the specific mRNA targets and biological processes regulated by DCP1a and DCP1b in humans.

Material and methods

DNA constructs

The DNA constructs used in this study are listed in **Supplementary Table 1**. The plasmids for expression of the β -globin-6xMS2bs and the control β -globin-GAPDH (control) mRNAs were kindly provided by Dr. J. Lykke-Andersen and have been described previously (Lykke-Andersen *et al.*, 2000 [DOI](#)).

Cell culture and transfections

Human HEK-293T cells were purchased from ATCC (CRL-3216) and were cultured in DMEM (Gibco 11995) supplemented with 10% (v/v) fetal bovine serum (Gibco) and grown at 37 °C with 5% CO₂. The identity of these cells was authenticated through SRT profiling, and they were confirmed to be negative for mycoplasma.

CRISPR/Cas9-mediated gene editing

The HEK-293T DCP1a-, DCP1b-, and DCP1a/b-null cell lines were generated by CRISPR/Cas9-mediated gene editing as described previously (Sgrono *et al.*, 2018 [DOI](#)). The guide RNA targeting human DCP1 paralog genes (DCP1a: GCTGTTTCGAGGTAAGCTGG; DCP1b:

TCGGCCATCGGGCCAACGAG) was designed using the ATUM CRISPR gRNA Design tool. Genomic DNA was isolated from single clones using the Wizard SV Genomic DNA Purification System (Promega). The DCP1a/DCP1b locus was PCR amplified, and Sanger sequencing of the targeted genomic regions confirmed the presence of a deletion within the targeted exon resulting in a frameshift. The absence of detectable levels of DCP1a or DCP1b protein was confirmed by Western blotting.

Tethering assays

Tethering assays using the MS2 reporter system were performed as described previously (Chang *et al.*, 2014 [DOI](#)). Briefly, HEK-293T cells were cultured in six-well plates and transiently transfected with a mixture of four plasmids: 0.5 µg control plasmid (β-globin-GAPDH), 0.5 µg plasmid encoding the β-globin-6xMS2bs, 0.5 µg plasmids encoding the MS2-tagged fusion protein (SMG7, TNRC6A-SD (silencing domain), or Nanos1) and plasmids encoding GFP-MBP, GFP-DCP2 E148Q, or GFP-DCP1 (wild-type or fragments) as indicated amounts. The cells were harvested two days after transfection. Total RNA was isolated using the TRIzol method (Thermo), separated on a denaturing agarose gel, transferred onto a nylon membrane (Amersham Hybond-N⁺), and hybridized with a probe synthesized *in vitro* using the full-length β-globin DNA sequence labeled with ³²P. The hybridized membrane was then analyzed by Northern blot.

5'-phosphate-dependent exonuclease assay

The integrity of the 7-methylguanosine cap structure at the 5'-ends of the transcripts was investigated in a 5'-phosphate-dependent exonuclease assay. 10 µg RNA extracted from the specific tethering assays was incubated in a 20 µl reaction volume with 1 unit of Terminator 5'-phosphate-dependent exonuclease (Epicentre) for 60 min at 30°C. The Terminator was omitted in the control. The reaction was then stopped by adding phenol. After standard extraction and ethanol precipitation, RNA levels were visualized by northern blotting.

Immunofluorescence

Cells were fixed and permeabilized as described previously (Jakymiw *et al.*, 2005 [DOI](#)). The antibodies used in this study are listed in **Supplementary Table 2**.

Decapping assays

Decapping assays were performed with immunoprecipitated wildtype GFP-DCP2 or a catalytic mutant (E148Q) from either wild-type DCP1a/b-null HEK-293T cells, along with *in vitro*-synthesized RNA (127 nucleotides) with a ³²P-labeled cap structure (Chang *et al.*, 2014 [DOI](#)). To generate the ³²P-labeled cap RNA, we utilized the ScriptCap m⁷G Capping System and ScriptCap 2'-O-methyltransferase kit (Epicentre Biotechnologies) with [α-³²P] GTP for our experiments. The concentration of RNA used was 0.5 µM. The decapping reactions were performed at 30 °C for the indicated times in a total volume of 10 µl of decapping buffer (50 mM Tris-HCl (pH 7.5), 50 mM ammonium sulfate, 0.1% (w/v) BSA and 5 mM MgCl₂). The proteins were diluted to their working concentrations with the decapping buffer. The reactions were stopped by adding up to 50 mM EDTA (final concentration), and 1 µl of each sample was spotted on polyethylenimine (PEI) cellulose thin-layer chromatography plates (Merck) and developed in 0.75 M LiCl.

Co-immunoprecipitation assays and Western blot analysis

For immunoprecipitation assays, HEK-293T cells were seeded in 10 cm dishes and transfected with 20 µg total plasmid DNA using Lipofectamine 2000 (Thermo). The cells were washed 48 h after transfection with phosphate-buffered saline and lysed in 1 ml NET buffer (50 mM Tris-HCl (pH 7.5), 150 mM NaCl, 1 mM EDTA, 0.1% (v/v) Triton-X-100, 10% (v/v) glycerol and supplemented with complete protease inhibitor cocktail (Sigma)). Immunoprecipitations were performed as described previously (Braun *et al.*, 2012 [DOI](#)). RNase A (1 µl, 10 mg/ml) was added to the cell extract in all

experiments. The antibodies used in this study are listed in **Supplementary Table 2**. All Western blots were developed with the ECL Western blotting analysis system (GE Healthcare) as recommended by the manufacturer.

PNRC1 and PNRC2 knockdown

Plasmids expressing short-hairpin RNAs (shRNAs) for knockdowns were derived from the pSUPER plasmid containing the puromycin-resistance gene for selection. The vector backbone was a kind gift from O. Mühlemann (University of Bern). The 19 nt target sequences were as follows: control: ATTCTCCGAACGTGTACAG, PNRC1: CAAAGTTTAGTGATCCACCTTT, and PNRC2: AGTTGGAATTCTAGCTTAT. HEK-293T cells were grown in DMEM supplemented with 10% heat inactivated fetal bovine serum and 2 mM L-Glutamine. The cells were transfected in 6-well plates using Lipofectamine 2000 (Thermo) according to the manufacturers' protocol. Transfection mixtures contained 0.5 µg control plasmid (β-globin-GAPDH), 0.5 µg plasmid encoding the β-globin-6xMS2bs, 0.5 µg plasmids encoding the MS2-tagged SMG7, and 2.5 µg of plasmids expressing the relevant shRNA. Twenty-four hours after transfection, cells were selected in medium supplemented with 1.5 µg/ml puromycin. After 24 h of selection, cells were counted and reseeded in new 6-well plates in medium without puromycin for recovery. After 24 h of reseeded, the cells were prepared for re-transfection to conduct tethering assays.

RNA Granule Quantification

Quantification of P-bodies was performed using Fiji/ImageJ software. Nuclei were counted in the DAPI channel utilizing the cell counter plugin. P-bodies in the 488 channel (stained with EDC4 for P-bodies) were detected using a binary threshold and primary object detection for objects between 40 and 255 pixels. P-body sizes were quantified using the "Analyze Particles" function, with detection parameters set to sizes between 0.00 and infinity (inch²) and circularity between 0.00 and 1.00. Size and intensity thresholds were maintained consistently across experiments with identical staining parameters. Excluding P-bodies on image edges, all results are presented as means ± standard errors of the means (SEMs) from experiments repeated independently at least three times. Unpaired t-tests were used to evaluate statistical differences between samples. Quantification of P-body size in wild-type and DCP1a/b-null HEK-293T cells was conducted, measuring average granule size across at least three fields of view. The middle line represents the mean of these measurements, with P-values calculated using unpaired t-tests (*: $p \leq 0.05$; **: $p \leq 0.01$; ***: $p \leq 0.001$).

RNA immunoprecipitation assays

To perform RNA immunoprecipitation assays, HEK-293T DCP1a/b-null cells were cultured in 10 cm dishes and transiently transfected with a mixture of four plasmids: 5 µg of β-globin-6xMS2bs, 5 µg of plasmids encoding the MS2-HA-Strep-tagged SMG7, 5 µg of plasmids encoding V5-Streptavidin-Binding Peptide (SBP)-tagged DCP2 E148Q, and 5 µg of plasmids encoding GFP-MBP. Forty-eight hours after transfection, V5-SBP-DCP2 E148Q or MS2-HA-Strep-SMG7 was immunoprecipitated by Streptavidin (Thermo) or Strep-Tactin (IBA) beads, respectively, as described previously (Braun *et al.*, 2012 [DOI](#)).

The level of DCP2-bound reporter mRNA was extracted and reverse transcribed using β-globin reverse primer (TTAGTGATACTGTGGGCCAGGGC). The amount of target mRNA was subsequently determined using quantitative PCR (qPCR) using the respective primer pair (Fw: ATGGTGCACCTGACTCCTGAG; Rev: TTAGTGATACTGTGGGCCAGGGC). To use the Livak method ($\Delta\Delta CT$ method) for relative quantification, each primer pair was tested if the amplification rate of the specific PCR product was $2 \pm 5\%$. To determine the level of DCP2 bound to reporter mRNA, Strep-tagged SMG7 was precipitated using Strep-Tactin beads, and the DCP2 level was detected by Western blot.

Transcriptome sequencing (RNA-Seq)

HEK-293T wild-type, DCP1a-, DCP1b-, or DCP1a/b-null cells were plated on 15 cm dishes 24 h before harvesting as described previously (Calviello et al., 2016 [DOI](#)). Total RNA was extracted using the RNeasy Mini Kit (Qiagen) and a library prepared using the TruSeq RNA Sample Prep Kit (Illumina). Three biological replicates were analysed. RNA-Seq libraries were sequenced using the Illumina NovaSeq 6000 sequencing system.

Metabolomics analysis (HILIC, LC-QToF)

LC-MS grade methanol and water were procured from Scharlau Chemie (Sentmenat, Barcelona, Spain) and MS-grade acetonitrile was obtained from J.T. Baker (Phillipsburg, NJ, USA). The remaining chemicals were purchased from MilliporeSigma (St. Louis, MO, USA), unless specified otherwise.

The number of cultured cells in each tube was adjusted to 10^6 in 100 μ l PBS and quenched by using 400 μ l of cold methanol. After centrifuging at 15000 $\times g$ for 5 min, the supernatant was collected and stored at -20°C until use. Metabolomics data were acquired using an Agilent 1290 ultra-high-performance liquid chromatography (UHPLC) system (Agilent Technologies, Waldbronn, Germany) connected to a Bruker maXis ultra-high-resolution (UHR)-time-of-flight (TOF) mass spectrometer (Bruker Daltonics, Bremen, Germany). Metabolites were separated using a BEH Amide column (2.1 mm $\times$ 100 mm, 1.7 μ m), with an injection volume of 10 μ l. The autosampler and column temperatures were maintained at 4 and 40 $^{\circ}\text{C}$, respectively. The mobile phase A consisted of 10 mM ammonium acetate in deionized water with 0.1% formic acid, and the mobile phase B consisted of 10 mM ammonium acetate in a 5:95 (v/v) water:acetonitrile mixture with 0.1% formic acid. The flow rate was 0.4 mL min^{-1} . The elution gradient was as follows: 0–0.5 min, 99% mobile phase B; 0.5–7 min, 99%–50% mobile phase B; 7–10 min, 50% mobile phase B; and column re-equilibration with 99% mobile phase B for 2 min. The electrospray settings were: dry gas temperature, 200 $^{\circ}\text{C}$; dry gas flow rate, 8 L min^{-1} ; nebulizer gas pressure, 2 bar; capillary voltage, 4500 V; endplate offset potential, 500 V. Mass spectra were recorded over the range 50 to 1500 m/z in the positive mode. The TOF mass analyser was calibrated using sodium formate with a mass range of 50–1500 m/z before use.

The maXis UHR-TOF data were processed using MS-DIAL version 4.60 (accessible at <http://prime.psc.riken.jp/> [DOI](#)) and Bruker Compass DataAnalysis version 4.1. For peak detection and alignment, the minimum peak height was set at an amplitude of 3000, the retention time tolerance was set at 0.1 min, and other parameters were set at their default values using the web-based metabolomics data processing tool MetaboAnalyst 5.0 (accessible at <https://www.metaboanalyst.ca/> [DOI](#)). For statistical analysis, the data were normalized (mean-centred and divided by the standard deviation of each variable) and heat maps and volcano plots were constructed. Partial least squares discriminant analysis (PLS-DA) was performed using the R package, mixOmics (Rohart et al., 2017 [DOI](#)). Known features were first confirmed using a home-built compound library generated using commercial reference standards based on retention time and isotopic mass information. Unknown features of interest were further identified using the online databases METLIN (Guijas et al., 2018 [DOI](#)) and the Human Metabolome Database (Wishart et al., 2018 [DOI](#)).

Transcriptomics analysis

Transcriptomes were sequenced using high-throughput technology. Lo-quality bases and adapters were removed with Trimmomatic (Bolger et al., 2014 [DOI](#)). The filtered, high-quality reads were mapped to the human genome (GENCODE Human Release 33) and quantified with RSEM (Frankish et al., 2019 [DOI](#); Li and Dewey, 2011 [DOI](#)). The TPM expression profiles from each sample were then used to generate PCA plots with R. The fold changes between different groups for each gene were estimated with DESeq2 (Love et al., 2014 [DOI](#)). Genes were defined as differentially expressed if the

absolute value of the corresponding fold change was larger than 1.5 and the adjusted p value was smaller than 0.05. Pathway analysis was performed using the clusterProfiler R package (Wu et al., 2021b [↗](#)); specifically, GSEA (Subramanian et al., 2005 [↗](#)) was used to identify activated or suppressed GO terms (The Gene Ontology, 2019), REACTOME pathways (Croft et al., 2011 [↗](#)) and KEGG pathways (Kanehisa and Goto, 2000 [↗](#)). All the plots were generated with in-house R scripts using the packages ggplot2 and ComplexHeatmap (Gu et al., 2016 [↗](#); Wickham, 2016 [↗](#)). The expression levels of DCP1a and DCP1b for different cancer types in TCGA were download from PanCanAtlas (Cancer Genome Atlas Research et al., 2013). Survival analysis were carried out with webserver DoSurvive (Wu et al., 2023 [↗](#)).

Results

Knockout of human DCP1 represses the general mRNA decapping process *in vivo*

To investigate the roles of the two DCP1 paralogs, we knocked out DCP1a, DCP1b, or both in HEK-293T cells using the CRISPR-Cas9 technique. Western blot analysis and Sanger sequencing confirmed the successful generation of the knockout cell lines (Supplementary Figure S1A-C). We further investigated whether the absence of DCP1a, DCP1b, or both affected decapping. To monitor mRNA decay, the NMD factor SMG7 was tethered to an mRNA reporter in the DCP1a/b-knockout cells (**Figure 1A** [↗](#)). While DCP1 is the main decapping activator in yeast, previous studies in metazoans have shown that at least two decapping activators have to be simultaneously depleted to effectively inhibit mRNA decapping (Braun et al., 2012 [↗](#); Sakuno et al., 2004 [↗](#)). Surprisingly, we observed that the depletion of both DCP1a and DCP1b substantially inhibited SMG7-mediated mRNA degradation, resulting in the accumulation of deadenylated reporter RNA molecules (**Figure 1B** [↗](#), lane 2 and 6).

To investigate whether the inhibitory effect of DCP1 knockout on decapping is specific to SMG7 or a general effect, we performed tethering assays using either the mRNA decay factor Nanos1 or the crucial miRNA component TNRC6A to induce 5'-3' mRNA decay. In both cases, DCP1 knockout led to the substantial accumulation of mRNA decay intermediates (Supplementary Figure S2A-C), indicating that DCP1 is important for the decapping and degradation of mRNA in general.

To confirm the presence of a cap structure on the accumulated reporter RNA intermediate, we treated it with Terminator 5'-phosphate exonuclease. The resistance of the transcripts to this treatment is strong evidence that the accumulated RNA fragments had intact 5' caps (Figure. S2D). The accumulation of capped mRNA decay intermediates suggests slowed or impaired DCP2 mediated decapping in the absence of DCP1. Taken together, these findings provide compelling evidence for the important role of DCP1 in the decapping process in human cells, mirroring its established role in yeast.

To better understand the functional roles of the DCP1 paralogs in decapping, we performed tethering assays and reintroduced GFP-tagged DCP1a or DCP1b into DCP1a/b-knockout cells, leading in both cases to the restoration of decapping and degradation of the reporter mRNA (**Figure 1C, D** [↗](#); Supplementary Figure S2E). This finding strongly suggests that DCP1a and DCP1b are functionally redundant for the general decapping of mRNA.

DCP1 EVH1 domain is crucial for cellular mRNA decapping process

Building on our finding that DCP1 is crucial for efficient mRNA decapping in human cells, we sought to identify the specific region of DCP1 that is important for the function of the decapping complex. Given the observed functional redundancy of DCP1a and DCP1b in the decapping process, we chose to use DCP1a as a model (**Figure 1E** [↗](#)). In DCP1a/b knockout cells, we

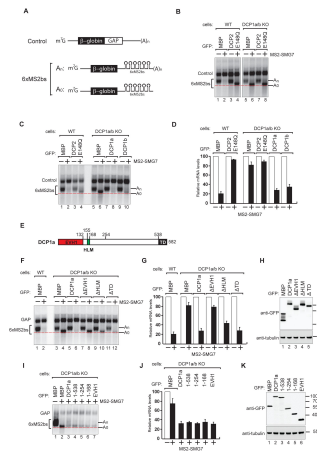


Figure 1

DCP1 is essential for human decapping process.

(A) Schematic of the reporters used in tethering assays.

(B) Wild-type or DCP1a/b knockout HEK-293T cells were transfected with a mixture of plasmids. One plasmid expressed the β -globin-6xMS2bs, and another plasmid expressed the transfection control, which contained the β -globin gene fused to the GAPDH 3' UTR, but it lacked the MS2 binding sites (β -globin-GAP, Control). The third plasmid expressed MS2-HA or MS2-tagged SMG7 proteins, and a fourth plasmid encoding a GFP-tagged protein was included in the transfection mixtures where indicated. A northern blot of representative RNA samples is shown. The positions of the polyadenylated (An) and the deadenylated (A0) forms of the β -globin-6xMS2bs reporters are indicated on the right. A red dotted line additionally marks the fast migrating deadenylated (A0) form. The DCP2 inactive mutant (E148Q) serves as a negative control, highlighting the importance of DCP1 in decapping by showing the absence of decapping activity without functional DCP2. All experimental results were independently repeated at least three times.

(C) Complementation assays with GFP-DCP1a or GFP-DCP1b constructs in HEK-293T DCP1a/b-null cells were performed similarly to panel (A). All experimental results were independently repeated at least three times. This experiment demonstrates whether the reintroduction of DCP1a or DCP1b can restore the decapping activity in cells lacking both DCP1 isoforms.

(D) The β -globin-6xMS2bs mRNA levels were normalized to those of the control mRNA. These normalized values were set to 100 in cells expressing MS2-HA (white bars). The mean values for relative mRNA levels in cells expressing MS2-SMG7 were estimated with standard deviations (SD) from three independent experiments (black bars). This quantification provides a comparative measure of the mRNA stability under different conditions.

(E) The domain organization of human DCP1a.

(F) Complementation assays with GFP-DCP1a deletion constructs in HEK-293T DCP1a/b-null cells performed similarly to panel (A). All experimental results were independently repeated at least three times. This experiment helps identify which domains of DCP1a are crucial for its function in the decapping process.

(G) The β -globin-6xMS2bs mRNA levels were normalized to those of the control mRNA. These normalized values were set to 100 in cells expressing MS2-HA (white bars). The mean values for relative mRNA levels in cells expressing MS2-SMG7 were estimated with standard deviations (SD) from three independent experiments (black bars). This data provides insight into the functional importance of different DCP1a domains.

(H) A western blot demonstrating equivalent expression of the GFP-tagged proteins in panel (F). Tubulin served as a loading control, confirming that the expression levels of the deletion constructs were comparable.

(I) Complementation assays with GFP-DCP1a fragment constructs in HEK-293T DCP1a/b-null cells performed similarly to panel (A). All experimental results were independently repeated at least three times. This set of experiments further delineates the specific regions of DCP1a necessary for its decapping activity.

(J) The β -globin-6xMS2bs mRNA levels were normalized to those of the control mRNA. These normalized values were set to 100 in cells expressing MS2-HA (white bars). The mean values for relative mRNA levels in cells expressing MS2-SMG7 were estimated with standard deviations (SD) from three independent experiments (black bars). This helps validate the findings regarding the essential regions of DCP1a.

(K) A western blot demonstrating equivalent expression of the GFP-tagged proteins in panel (I). Tubulin served as a loading control, ensuring that the variations in mRNA levels were due to functional differences in the DCP1a fragments rather than differences in protein expression.

overexpressed DCP1a fragments lacking either of three major domains (EVH1, HLM, TD) and monitored mRNA degradation status (**Figure 1F-H**). Strikingly, we observed that overexpression of a DCP1a fragment lacking EVH1 domain (Δ EVH1) could not rescue the decapping defect, indicating that this domain is crucial to coordinate the cellular decapping process. Overexpression of DCP1a without the HLM domain (Δ HLM) partially restored the decapping defect, indicating a cooperative role for this domain in coordinating DCP1's decapping function. Adequate mRNA degradation occurred in cells overexpressing DCP1a fragment without TD domain.

To precisely identify the minimal functional region of DCP1, we generated a series of DCP1a expression constructs by sequentially removing its domains. The structured core of DCP1 comprises its N-terminal EVH1 and HLM domains, followed by a long unstructured region (amino acids 169-538) and the terminal trimeric domain (TD). Given the presence of this unstructured region, we utilized AlphaFold to predict areas potentially possessing secondary or tertiary structures. This approach enabled us to design more effective truncations. We began by referencing an AlphaFold prediction indicating that the DCP1a segment spanning amino acids 1-254 is structured (AlphaFoldDB: A0A087WT55). Using these predictions, we systematically deleted specific domains and regions. This stepwise truncation method allowed us to pinpoint the crucial segments required for DCP1a's function with greater accuracy. Overexpression and subsequent complementation assays in DCP1a/b-knockout cells revealed that all the constructs containing the EVH1 domain successfully restored the decapping ability of the enzyme complex (**Figure 1I-K**). Together, these findings indicate that the EVH1 domain of DCP1 is essential for the formation of a functional decapping complex.

DCP1 knockout does not affect the formation of P-bodies or the enzymatic activity of DCP2

Since we observed that DCP1 appears to be indispensable for mRNA decapping, we sought to identify the precise role of DCP1 in decapping. To begin with, we explored P-body formation and DCP2 localization in DCP1a/b-knockout cells. Immunofluorescence and confocal microscopy analysis of P-body formation indicated that DCP1 has little involvement in P-body formation. In DCP1a/b-knockout cells, DCP2 remained localized in P-body granules, indicating no correlation between DCP1 and P-body formation or DCP2 localization (**Figure 2A**; Supplementary Figure S3A, B). Meanwhile, we observed that in DCP1a/b knockout cells, although the number of P-bodies only slightly increased, their size significantly enlarged (**Figure 2B**; Supplementary Figure S3C). This suggests that the knockout of DCP1a/b, while not affecting P-body formation, still has some impact on the morphology of P-bodies.

Next, we wanted to clarify if the absence of cellular DCP1 has a direct effect on decapping activity of DCP2. To evaluate the effects of DCP1 on the decapping activity of DCP2, we performed *in vitro* decapping assays by immunoprecipitating DCP2 extracted from wild-type and DCP1a/b-knockout cells. We observed a decrease in the enzymatic activity of DCP2 immunopurified from DCP1a/b knockout cells, but it still retains a considerable decapping ability (**Figure 2C, D**; Supplementary Figure S3D). These data suggest that, in accordance with prior literature, DCP2 is sufficient for *in vitro* decapping without DCP1 (Lykke-Andersen, 2002; van Dijk *et al.*, 2002; Wang *et al.*, 2002). Nevertheless, our previous observations emphasize the significant role of DCP1a/b gene knockout in the regulation of mRNA decapping *in vivo*.

Although DCP2 still exhibits decapping activity *in vitro*, the decapping process of DCP2 is significantly inhibited *in vivo*. It should be noted that *in vivo* and *in vitro* experiments inherently measure different aspects of molecular function. The discrepancies between *in vivo* and *in vitro* experiments are of great significance because *in vivo* experiments more comprehensively reflect the regulatory role of DCP1. The *in vitro* environment lacks the complexity and numerous cofactors present *in vivo*, often failing to accurately replicate the interactions occurring within cells, highlighting the indispensable contribution of DCP1 in real mRNA decapping scenarios.

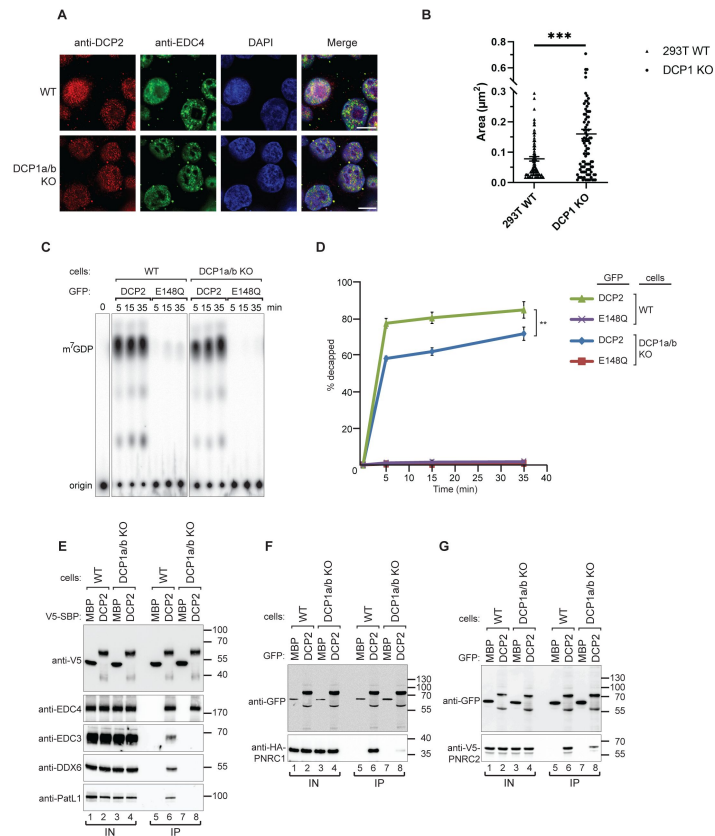


Figure 2

DCP1 serves as a bridging factor facilitating the interaction between multiple decapping factors and DCP2.

(A) HEK-293T wild-type or DCP1a/b-null cells were stained with antibodies detecting DCP2 (red) and a P-body marker, EDC4 (green), and then counterstained with DAPI to visualize the nucleus (blue). In the merged image, colocalization of DCP2 and EDC4 appears yellow, indicating their interaction in P-bodies. This staining helps to visualize the cellular localization of DCP2 in the presence or absence of DCP1a/b. Scale bar = 10 μm .

(B) Quantification of P-body size in wild-type and DCP1a/b-null HEK-293T cells. The average granule size was measured across at least three different fields of view. The middle line represents the mean of these measurements, and P-values were calculated using unpaired t-tests (ns: not significant; *: $p \leq 0.05$; **: $p \leq 0.01$; ***: $p \leq 0.001$).

(C) GFP-tagged DCP2 proteins were expressed in human HEK-293T wild-type or DCP1a/b-null cells. Following expression, the GFP-tagged DCP2 proteins were purified using GFP antibody and IgG beads. In vitro decapping activity was then tested, with the catalytically inactive DCP2 E148Q mutant serving as a negative control to demonstrate the specificity of the decapping activity. All experimental results were independently repeated at least three times.

(D) Decapping assays in vitro were conducted to measure the fraction of decapped mRNA substrate by detecting the release of m⁷GDP over time. The results are plotted as a function of time, with error bars representing the standard deviations (SD) from three independent experiments. An unpaired t-test was used to evaluate the statistical difference between samples (ns: not significant; *: $p \leq 0.05$; **: $p \leq 0.01$; ***: $p \leq 0.001$). This panel demonstrates the decapping efficiency of DCP2 in the presence or absence of DCP1a/b.

(E) V5-Streptavidin-Binding Peptide (SBP)-DCP2 proteins were expressed in human HEK-293T wild-type or DCP1a/b-null cells, followed by purification using Streptavidin beads. This experiment examines the interaction of V5-SBP-tagged DCP2 with endogenous decapping factors in wild-type or DCP1a/b knockout HEK-293T cells. The bound proteins were detected via Western blot, with V5-SBP-MBP employed as a negative control. All experimental results were independently repeated at least three times. This panel highlights the role of DCP1a/b in facilitating or stabilizing interactions between DCP2 and other decapping factors.

(F-G) The interaction of GFP-tagged DCP2 with HA-tagged PNRC1 (F) or V5-tagged PNRC2 (G) was assessed. The proteins were immunoprecipitated using anti-GFP antibodies and analyzed by Western blotting with the indicated antibodies. All experimental results were independently repeated at least three times. These panels demonstrate the interaction between DCP2 and specific decapping co-factors, indicating how DCP1a/b may influence these interactions.

Therefore, our study indicates that relying solely on *in vitro* experiments is insufficient to fully understand the function of DCP1; while important, *in vitro* experiments cannot completely capture the dynamics within cells. Together, these experiments deepen the understanding of DCP1's role in the decapping process *in vivo*. The lack of a significant difference in decapping activity observed *in vitro* does not negate DCP1's regulatory role in the cellular context; rather, it underscores our previous oversight of DCP1's importance under *in vitro* conditions and the fact that DCP1's regulatory mechanisms in the decapping process remain to be elucidated.

DCP1 is a central scaffold protein that recruits multiple decapping factors to DCP2

Since the discrepancies between *in vivo* and *in vitro* experiments reflect the regulatory role of DCP1 in decapping machinery. There are various factors and complex regulatory networks in the *in vivo* environment, which make it challenging to accurately simulate *in vitro*. To determine the underlying cause, we postulated that the complexity of the cellular setting, where multiple factors and cofactors are at play, contributes to these discrepancies. The observed cellular decapping defect in DCP1a/b knockout cells might be attributed to DCP1 functioning as a scaffold. It is conceivable that DCP1 facilitates the assembly of decapping modules, enabling their interaction with DCP2. Alternatively, this discrepancy could stem from the unique contributions of individual subunits, such as EDC4, EDC3, DDX6, and other decapping factors, to DCP2's activity. To explore this possibility, our investigation aimed to identify the specific decapping factors that rely on DCP1 for interaction with DCP2. We overexpressed V5-SBP-or GFP-tagged DCP2 in wild-type and DCP1a/b-knockout cells and then performed coimmunoprecipitation assays with RNase A treatment. In the absence of DCP1, interactions between DCP2 and the decapping factors EDC3, DDX6, and PatL1 (Edc3, Dhh1, and Pat1 in yeast, respectively) were almost completely suppressed, but not with EDC4 (also known as Ge-1 in *Drosophila*), a major component of the core decapping complex in metazoans (Figure 2E [↗](#)).

Furthermore, the interaction between DCP2 and PNRC1, as well as PNRC2, which acts in synergy with DCP1 to promote decapping, was reduced in DCP1a/b-knockout cells (Figure 2F, G [↗](#)). These findings suggest that DCP1 acts as a binding platform for multiple decapping factors and is essential for the interaction of these cofactors with DCP2 in human cells.

DCP2-mediated decapping is not merely due to the presence of decapping factors on the DCP1a scaffold

Based on our observation that DCP1a EVH1 domain is sufficient to rescue the decapping defect in DCP1a/b knockout cells, we hypothesized that the DCP1 EVH1 domain may play a key role in regulating the decapping machinery. To evaluate the contribution of this domain to the formation of the decapping complex, we sought to identify which decapping factors interact with the DCP1 EVH1 domain. We overexpressed full-length or fragmented GFP-tagged DCP1a in DCP1a/b-knockout cells and subsequently immunoprecipitated the protein in the presence of RNase A. The EVH1 domain of DCP1a did not interact with any of the decapping factors except for DCP2 and PNRC1/2 (Figure. 3A-D [↗](#)).

PNRC1 has been shown to interact with the NMD factor UPF1, indicating that it is involved in the NMD pathway (Cho *et al.*, 2009 [↗](#)). Furthermore, it has recently been shown that PNRC1 serves as a regulator of rRNA maturation within nucleoli, tightly controlling the process of ribosomal RNA maturation (Gaviraghi *et al.*, 2018 [↗](#)). Previous studies have also demonstrated the involvement of PNRC2 in the NMD pathway. Structural analysis of the PNRC2-DCP1a complex has shown that PNRC2 can directly activate the decapping process (Cho *et al.*, 2013 [↗](#); Cho *et al.*, 2009 [↗](#); Lai *et al.*, 2012 [↗](#)). Given this background, we investigated the functional role of PNRC1/2 in modulating the activity of DCP2 through DCP1 EVH1 domain.

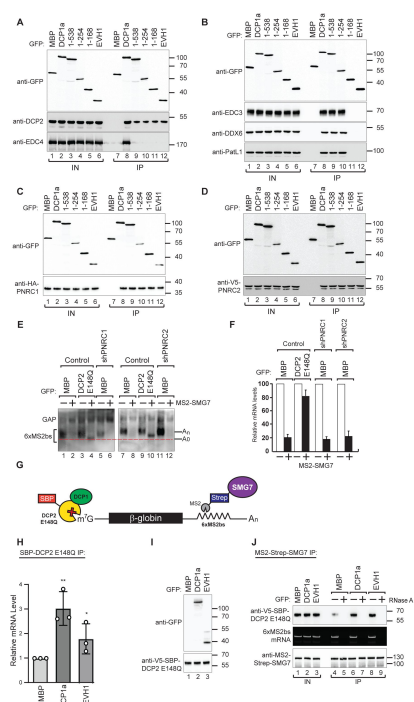


Figure 3

DCP1 facilitates DCP2 interactions with RNA molecules in human cells.

(A-D) The interaction of GFP-tagged DCP2 with various decapping factors was examined. Proteins were immunoprecipitated using anti-GFP antibodies and analyzed by Western blotting as described in **Figures 2D** and **2E**. All experimental results were independently repeated at least three times. These panels demonstrate the binding affinity and specificity of DCP2 for different decapping co-factors, highlighting the importance of these interactions in the decapping process.

(E) Tethering assays were performed in HEK-293T cells using control, shPNRC1, or shPNRC2 plasmids, following a procedure similar to that described in **Figure 1A**. All experimental results were independently repeated at least three times. This panel investigates the effect of knocking down PNRC1 or PNRC2 on the decapping activity, showing how these factors contribute to the stability and regulation of the β -globin-6xMS2bs mRNA.

(F) The β -globin-6xMS2bs mRNA levels were normalized to those of the control mRNA. These normalized values were set to 100 in cells expressing MS2-HA (white bars). The mean values for relative mRNA levels in cells expressing MS2-SMG7 were estimated with standard deviations (SD) from three independent experiments (black bars). This quantification provides a comparative measure of the mRNA stability under different experimental conditions.

(G) Schematic representation of the experimental procedure used in panels H and J. This diagram outlines the steps and components involved in the transfection and subsequent analysis, providing a visual aid for understanding the experimental setup.

(H) DCP1a/b-null HEK-293T cells were transfected with a mixture of plasmids, including β -globin-6xMS2bs, MS2-HA-Strep-SMG7, V5-SBP-DCP2 E148Q, and either full-length or the EVH1 domain of GFP-DCP1a. GFP-MBP was used as a control. The levels of β -globin-6xMS2bs mRNA bound to V5-SBP-DCP2 E148Q were immunoprecipitated using streptavidin beads and quantified by RT-PCR with GAPDH as a reference. The results represent three biological replicates' mean values $\pm$ standard deviations (SD). An unpaired t-test was used to evaluate the statistical difference between samples (ns: not significant; *: $p \leq 0.05$; **: $p \leq 0.01$; ***: $p \leq 0.001$). This experiment assesses the role of different DCP1a constructs in the interaction and binding efficiency of DCP2 to the target mRNA.

(I) The V5-SBP-DCP2 E148Q immunoprecipitated samples corresponding to panel H were analyzed by Western blotting using the indicated antibodies. This panel confirms the tagged proteins' expression and proper immunoprecipitation, ensuring the RT-PCR results' validity in panel H.

(J) The plasmid mixture described in panel H was transfected into DCP1a/b-null HEK-293T cells. Subsequently, Strep-tag beads were used to immunoprecipitate the MS2-HA-Strep-SMG7 bound β -globin-6xMS2bs mRNA to study the *in vivo* interaction levels between RNA molecules and DCP2 E148Q in the presence of either full-length GFP-DCP1a or its EVH1 domain. GFP-MBP served as a control. All experimental results were independently repeated at least three times. This panel explores how different domains of DCP1a influence the interaction between DCP2 and its RNA targets, providing insights into the functional domains required for effective decapping.

To determine whether PNRC1 and PNRC2 are necessary for DCP2 activity *in vivo*, we used shRNA to knockdown the expression of PNRC1 and PNRC2 individually (Supplemental Figure S4). Unlike the DCP1 knockout, the knockdown of PNRC1 or PNRC2 had only a minor effect on the results of the tethering assays, suggesting that PNRC1 and PNRC2 are not essential for DCP2 activity (**Figure 3E, F**). Therefore, we concluded the activation of the decapping complex by the DCP1a cannot be solely attributed to the binding of decapping factors on the DCP1a scaffold. This observation suggests that while DCP1's scaffolding function is crucial for recruiting cofactors, the decapping process likely involves additional layers of regulation that are not fully understood.

DCP1 increases the mRNA-binding affinity of DCP2

Our findings suggest that the marked reduction in decapping activity observed in DCP1a/b-knockout cells is not exclusively essential for DCP1's function as a scaffold in the decapping process. This discrepancy prompts an investigation into the potential role of the DCP1 EVH1 domain, as previous studies have theorized its critical involvement in DCP2 mRNA binding (Chang *et al.*, 2014; Valkov *et al.*, 2016), although this aspect has not been explored to date. To evaluate this hypothesis, we investigated the molecular mechanisms of DCP1-mediated substrate binding by DCP2.

We evaluated the impact of DCP1 on interactions between DCP2 and mRNA in human cells using tethering assays with a catalytically inactive mutant DCP2 protein (DCP2 E148Q) introduced into DCP1a/b-knockout cells. This prevented the degradation of target mRNAs and maintained them in the decapping stage. Full-length DCP1a or the EVH1 domain alone were also introduced into the cells (**Figure 3G**). DCP2 samples were then immunoprecipitated and real-time PCR was used to measure the concentrations of reporter mRNAs that interacted with DCP2. Our results show that the mRNA binding affinity of DCP2 is weak in the absence of DCP1 but significantly increases in the presence of DCP1a full-length or the EVH1 domain (**Figure 3H, I**). While these changes may initially seem minor *in vitro*, their cumulative impact in the dynamic cellular environment could be substantial. Even minor perturbations in RNA binding affinity can trigger cascading effects, leading to significant changes in decapping activity and the accumulation of deadenylated intermediates upon DCP1 depletion. Cellular processes involve complex networks of interrelated events, and small molecular changes can result in amplified biological outcomes.

To further bolster the validity of these findings, we conducted RNA immunoprecipitation (RNA-IP) assays. Reporter RNA was isolated during the decapping process and consistent with the findings of the DCP2 immunoprecipitation assays, high levels of DCP2 were observed in the RNA immunoprecipitation assays when the DCP1a EVH1 domain was overexpressed (**Figure 3J**). These findings strongly underscore the significance of the EVH1 domain of DCP1a in enhancing interactions between DCP2 and mRNA, demonstrating how subtle molecular variations observed *in vitro* may translate into significant phenotypic outcomes within the complex cellular environment, emphasizing the critical regulatory role of DCP1a in the cellular decapping process. Taken together, we demonstrated that DCP1a can regulate DCP2's cellular decapping activity by enhancing DCP2's affinity to RNA, in addition to bridging the interactions of DCP2 with other decapping factors. This represents a pivotal molecular mechanism by which DCP1a exerts its regulatory control over the mRNA decapping process.

DCP1a and DCP1b regulate distinct endogenous downstream mRNA

The existence of two apparently redundant DCP1 paralogs, DCP1a and DCP1b, is intriguing, and raises the question whether they are truly redundant or in fact have distinct roles. Considering the low sequence similarity of DCP1a and DCP1b and since DCP1 was observed to enhance mRNA recognition by DCP2, we hypothesized that the DCP1 paralogs may contribute to the preferential regulation of specific mRNA targets. In other words, although DCP1a and DCP1b may be redundant in the general mRNA decapping process, they may differentially regulate specific mRNA targets.

We performed an RNA-Seq analysis of HEK-293T cells lacking DCP1a, DCP1b, or both and compared the results with those from wild-type cells (**Figure 4A** [↗](#); Supplementary Figure S5A-D). There were significant differences in the transcript profiles of the knockout cells. In DCP1a-, DCP1b-, and DCP1a/b-knockout cells, the expression of 604, 1,255, and 1,146 genes were significantly downregulated, while 504, 677, and 1,383 genes were significantly upregulated (Supplementary Table 3-5). Notably, only a subset of genes dysregulated in both DCP1a- and DCP1b-knockout cells showed alterations in DCP1a/b-knockout cells. This observation is further evidence of the crucial role of DCP1 in decapping (**Figure 4B** [↗](#)). It's worth mentioning that these significantly altered genes could be caused by both the direct effect of the accumulation of the target genes or the indirect effect of the accumulation of target genes.

Furthermore, comparing the transcriptomes of the DCP1a- and DCP1b-knockout cells to the wild type result in many genes that are uniquely altered DCP1a-knockout or DCP1b knockout (**Figure 4A** [↗](#) and **4B** [↗](#)), suggesting that these two proteins regulate distinct sets of genes, whether directly or indirectly. This observation implies that DCP1a and DCP1b are non-redundant in gene expression regulation.

DCP1a and DCP1b play distinct roles in cancer and gene expression regulation

To identify the gene sets associated with DCP1a and DCP1b, we performed GSEA for Hallmark, Gene Ontology (GO), Kyoto Encyclopedia of Genes and Genomes (KEGG), and REACTOM pathways using the transcriptome data from DCP1a and DCP1b knockout experiments. Analysis of the Hallmark pathways revealed that multiple cancer-related pathways were activated in DCP1a-knockout and/or DCP1b-knockout cells, such as apoptosis, E2F targets, MYC targets, the mTOR signalling pathway, the G2/M checkpoint pathway, the KRAS signalling pathway, and the epithelial-mesenchymal transition pathway (**Figure 4C** [↗](#)). Results for the GO, KEGG and REACTOM pathways likewise showed that DCP1a is involved in other cancer-related pathways, such as the Notch signalling pathway, embryonic skeletal system development, vasculature development, and the mitotic cell cycle pathway (**Figure 4D** [↗](#); Supplementary Figure S5E-H). These results are consistent with recent findings that DCP1a is strongly associated with embryonic growth and tumour development (Ibayashi et al., 2021 [↗](#); Wu et al., 2018 [↗](#); Wu et al., 2021a [↗](#)), and suggest that DCP1a is more strongly associated with cancer than is DCP1b.

Furthermore, we noted that the activation of capped intron-containing pre-mRNA processing was evident in both DCP1a- and DCP1b-deficient cells (Supplementary Figure S5H). This observation aligns with the understanding that the depletion of DCP1a not only diminishes mRNA degradation and transcription but also underscores the pivotal role of DCP1 in the orchestration of gene expression. To further assess the clinical relevance of our findings, we evaluated the expression levels of DCP1a and DCP1b using data from The Cancer Genome Atlas (TCGA) database (**Figure 4E** [↗](#)). Consistent with the above results, we found that expression levels of DCP1a were reduced in various cancers: bladder urothelial carcinoma, uterine corpus endometrial carcinoma, head and neck squamous cell carcinoma, prostate adenocarcinoma, cervical squamous cell carcinoma and endocervical adenocarcinoma, lung squamous cell carcinoma, kidney renal clear cell carcinoma, breast invasive carcinoma, thyroid carcinoma, lung adenocarcinoma, pheochromocytoma and paraganglioma, and thymoma. In contrast, DCP1b exhibited significant upregulation in certain cancer types while showing downregulation in others. Additionally, higher expression levels of DCP1a are found to be associated with a longer progression-free interval (PFI) in uterine carcinosarcoma carcinoma (Supplementary Figure S6A). Notably, higher expression of DCP1b is associated with a shorter PFI in brain lower grade glioma (Supplementary Figure S6B) and associated with a longer PFI in many other cancer types, including adrenocortical carcinoma, prostate adenocarcinoma, uveal melanoma, pancreatic adenocarcinoma, and uterine corpus endometrial carcinoma (Supplementary Figure S6C). This suggests that the role of DCP1b in tumorigenesis is multifaceted and context-dependent. In summary, these findings indicate that

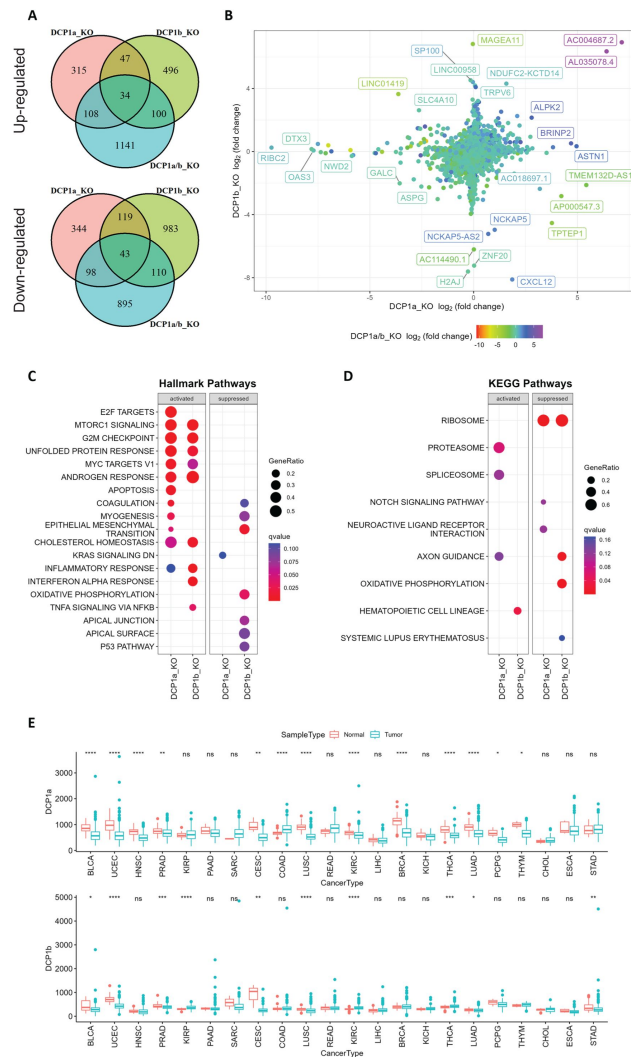


Figure 4

Gene expression analysis and pathway enrichment reveal distinct roles of DCP1a and DCP1b in human cells.

(A) The upper and lower panels display Venn diagrams illustrating the number of genes that were significantly up- and down-regulated in DCP1a, DCP1b, and DCP1a/b knockout cells (referred to as DCP1a_KO, DCP1b_KO, and DCP1a/b_KO), respectively. The overlapping regions indicate the number of genes significantly altered in multiple cell lines, highlighting shared and unique gene expression changes across different knockouts.

(B) The plot shows the distribution of fold changes in gene expression when comparing DCP1a_KO or DCP1b_KO to wild-type (WT) cells. The points are colored according to the fold changes observed when comparing DCP1a/b_KO to WT, visually representing how the loss of DCP1a and DCP1b individually and combined affects gene expression.

(C) Gene Set Enrichment Analysis (GSEA) results for hallmark pathways comparing DCP1a knockout and WT cells (left panel) and DCP1b knockout and WT cells (right panel). Activated and suppressed pathways are shown, with dots colored based on the q-value, indicating the statistical significance of the enrichment.

(D) GSEA results for KEGG pathways comparing DCP1a knockout and WT cells (left panel) and DCP1b knockout and WT cells (right panel). Like panel (C), activated and suppressed pathways are shown, with dots colored according to the q-value. This analysis helps identify the biological pathways most affected by the loss of DCP1a or DCP1b.

(E) Boxplots of the mRNA expression levels of DCP1a and DCP1b in various cancer types from The Cancer Genome Atlas (TCGA). Statistically significant differences in expression levels are labeled on top of each pair, with significance determined by the Wilcoxon test (ns: not significant; *: $p \leq 0.05$; **: $p \leq 0.01$; ***: $p \leq 0.001$). This panel provides insights into the potential roles of DCP1a and DCP1b in different cancers, highlighting their differential expression patterns.

DCP1a and DCP1b likely have distinct targets and play non-redundant roles. The nuanced functions of DCP1a and DCP1b warrant further investigation to fully elucidate their specific contributions.

DCP1a and DCP1b play different roles in cellular metabolism

To decipher the distinct biological functions of DCP1a and DCP1b, we employ metabolic profiling to investigate the complex metabolic alterations resulting from their absence, thus deepening our grasp of their cellular roles. Through untargeted metabolomic profiling of DCP1a-, DCP1b-, and DCP1a/b-knockout cells, we aim to thoroughly assess the impact of these knockouts, uncovering the unique roles of these proteins. After removing features that appeared in control samples and manually removing features with a poor peak shape and extremely low abundance, we identified approximately 123 metabolites in the experimental samples. We successfully differentiated DCP1a- and DCP1b-knockout cells using PLS-DA score plots, loading plots, and heatmaps of the metabolomic data and the differences in the metabolomic profiles revealed the distinct impacts of DCP1a and DCP1b on cellular metabolism (**Figure 5A** [↗](#); Supplementary Figure S7A, B). Additionally, the volcano plots illustrated the significant metabolic disparities among DCP1a-, DCP1b-, and DCP1a/b-knockout cells (**Figure 5B-D** [↗](#)), emphasizing the presence of noticeable differences. Remarkably, uridine diphosphate (UDP)-N-acetylglucosamine (GlcNAc), GlcNAc, glycerophosphoserine, and glycerophosphocholine levels were significantly elevated in DCP1a- and DCP1a/b-knockout cells, whereas those of nicotinamide adenine dinucleotide (NAD) were reduced. By contrast, DCP1b-knockout cells had elevated levels of lysophosphatidylethanolamine (P-18:0) [LPE (P-18:0)] (**Figure 5E, F** [↗](#); Supplementary Figure S7C). These results suggest that DCP1a and DCP1b have different effects on these metabolic pathways.

The significantly altered metabolites in the DCP1-knockout cells included UDP-GlcNAc and GlcNAc, which are associated with the amino sugar and nucleotide sugar metabolism pathway (hsa00520) in KEGG (Supplementary Figure S8). Further investigation revealed that the levels of these metabolites are correlated with the transcript levels associated with the amino sugar and nucleotide sugar metabolism pathway (**Figure 5G** [↗](#)). Together, our data strongly imply that DCP1a and DCP1b may exert regulatory influence over UDP-GlcNAc and GlcNAc through pathways that extend beyond the scope of amino sugar and nucleotide sugar metabolism. This indicates that these proteins play intricate and far-reaching roles in modulating cellular metabolism. Further studies are necessary to unravel the specific mechanisms and pathways by which DCP1a and DCP1b control these metabolites and to fully elucidate their roles in cellular metabolism.

Discussion

DCP1 is a key component in the mRNA decay pathway, crucial for the regulation of gene expression. Research across various species, from yeast to humans, has provided insights into the conservation and evolution of mRNA decapping mechanisms. Notably, DCP1 interacts with other decapping proteins such as DCP2, forming a decapping complex for mRNA degradation.

Additionally, the localization and functional analysis of DCP1 in different model organisms, as well as its importance in plant species, highlight its role in post-transcriptional gene expression regulation across species ([Lall et al., 2005](#) [↗](#); [Sakuno et al., 2004](#) [↗](#); [Xu et al., 2006](#) [↗](#)). However, in human cells, the specific biological functions of DCP1, especially the distinctions between its two paralogs, DCP1a and DCP1b, have not been fully elucidated. In this study, we found that DCP1 plays a key role in promoting mRNA decapping in human cells. By generating cell lines deficient in DCP1a, DCP1b, or both, our research emphasizes the importance of human DCP1 in the decapping process. We also verified that DCP1 is not only a key platform for assembling various mRNA decapping cofactors, but also established that the EVH1 domain of DCP1 is critical for the mRNA decapping process. Through RNA immunoprecipitation assays, we discovered that the interactions

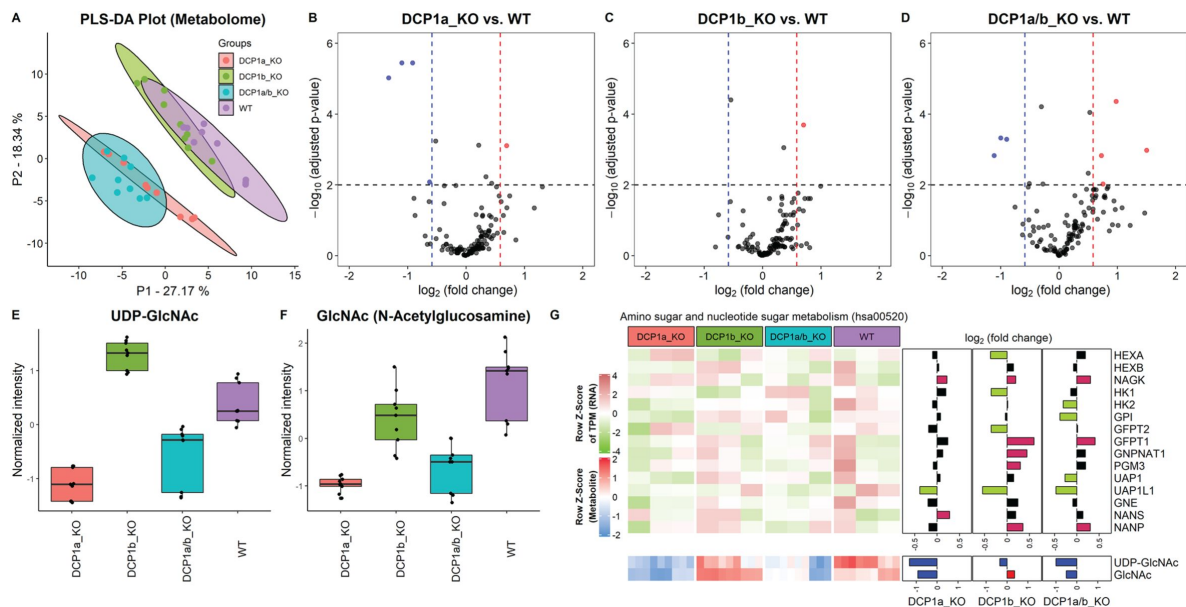


Figure 5

Metabolome profiling unveils unique functions of DCP1 paralogs in human cells.

(A) Partial Least Squares Discriminant Analysis (PLS-DA) plot illustrating the metabolome profiles of 12 samples, with 3 technical replicates for each knockout cell line. This plot shows how the metabolomic profiles of DCP1a_KO, DCP1b_KO, and DCP1a/b_KO cells differ from each other and wild-type (WT) cells, indicating distinct metabolic alterations resulting from the knockouts.

(B-D) Volcano plots comparing metabolite levels between (B) DCP1a_KO and WT, (C) DCP1b_KO and WT, and (D) DCP1a/b_KO and WT. Metabolites with an absolute fold change greater than 1.5 and a false discovery rate (FDR)-adjusted p-value less than 0.01 were considered significant. These plots highlight the metabolites significantly upregulated or downregulated in each knockout compared to WT, providing insight into the metabolic impact of DCP1a and DCP1b loss.

(E-F) Box plots of (E) UDP-GlcNAc and (F) GlcNAc levels between different cell lines. These plots show the relative abundance of specific metabolites in the different knockout and WT cells, indicating how the disruption of DCP1a and DCP1b affects specific metabolic pathways.

(G) Normalized expression levels, abundance of transcripts and metabolites, and fold changes in the amino sugar and nucleotide sugar metabolism pathway. Significantly upregulated and downregulated genes are colored in pink and blue, respectively. Upregulated and downregulated metabolites are colored in red and blue, respectively. This panel integrates transcriptomic and metabolomic data to provide a comprehensive view of how the amino sugar and nucleotide sugar metabolism pathway is altered in DCP1 knockout cells.

between DCP2 and mRNA heavily rely on the presence of DCP1's EVH1 domain. These findings imply that DCP1 facilitates the binding of DCP2 to target mRNAs through multiple mechanisms, thereby enhancing the efficiency of mRNA decapping. Finally, our integrated transcriptome and metabolome analyses comparing the specific functions of DCP1a and DCP1b in human cells revealed that they can regulate distinct endogenous mRNA targets and biological processes.

In a prior investigation, we made an intriguing discovery. In human cells, substituting the conserved NR-loop with a flexible GSSG loop in the EVH1 domain of DCP1 had no adverse effect on the interaction between DCP1 and DCP2, nor did it impede their integration into decapping factors. However, this seemingly subtle alteration had a significant consequence – it disrupted the activation of DCP2 (Chang *et al.*, 2014 [DOI](#)). This points to the pivotal role played by the EVH1 domain of DCP1 in the activation of DCP2, which, in turn, is a critical step in the mRNA decapping process. These findings, combined with our findings in the present study, provide compelling evidence that the DCP1 EVH1 domain is not just a passive component but a crucial player in coordinating interactions and facilitating conformational changes necessary for efficient mRNA decapping. A comprehensive functional characterization of these mechanisms is imperative to fully elucidate the mechanisms underlying DCP1-mediated activation of decapping.

While the protein components of the mRNA decapping machinery have been identified, our current understanding of the distinct roles played by individual decapping factors throughout the process remains limited. Notably, whether the different decapping factors confer target specificity has yet to be clarified. Although previous studies have suggested that DCP1a and DCP1b, the two paralogs of DCP1, play redundant roles in general mRNA decay, this notion is mainly based on reporter assays and single-gene approaches. The specific regulation of endogenous mRNA targets and biological processes by DCP1a and DCP1b is yet to be elucidated.

In this study, we investigated potential functional differences between DCP1a and DCP1b through a combined transcriptome and metabolome analysis. While DCP1a and DCP1b have both been implicated in cancer, our observation of significant alterations in multiple cancer-related pathways in DCP1a-knockout cells suggests that DCP1a might be more prominently associated with cancer development. Furthermore, the downregulation of DCP1a in multiple cancer types further supports the concept that DCP1a is a cancer-associated gene and suggests that its dysregulation may play a contributory role in the onset and progression of diverse cancers. (Ibayashi *et al.*, 2021 [DOI](#); Wu *et al.*, 2018 [DOI](#); Wu *et al.*, 2021a [DOI](#)). Survival analysis has indicated that elevated DCP1a expression in uterine carcinosarcoma carcinoma correlates with improved progression-free interval (PFI). Conversely, higher levels of DCP1b expression are associated with both favorable and unfavorable PFIs. Together with the observation of dysregulation of DCP1b in numerous cancers, we speculate that DCP1b is also implicated in cancer pathogenesis, and further studies are required to fully understand the roles of DCP1a and DCP1b in cancer pathogenesis.

With the transcriptome profiles of DCP1a and DCP1b knockout cells, we anticipate significant upregulation among potential targets. However, to discern direct targets from genes upregulated due to indirect effects, it is imperative to delve deeper into mRNA decay rates. Various tools and methods, such as time-series RNA-seq (Viegas *et al.*, 2023 [DOI](#)) or specialized sequencing techniques like Precision Run-On sequencing (PRO-seq) (Blumberg *et al.*, 2021 [DOI](#)) or RNA Approach to Equilibrium Sequencing (RATE-Seq) (Abdul-Rahman and Gresham, 2018 [DOI](#)), have been proposed to estimate mRNA decay rates. Luo *et al.* utilized TimeLapse-seq to identify potential targets for DCP2 (Luo *et al.*, 2021 [DOI](#); Schofield *et al.*, 2018 [DOI](#)). Incorporating these approaches could provide a more comprehensive understanding in this context.

Notably, our candidate target genes exhibited minimal overlap with the 91 targets of DCP2 (Luo *et al.*, 2021 [DOI](#)); only EPPK1 overlapped among potential targets of DCP1b_KO, while TES, PAX6, and C18orf21 were the only overlapping genes in the DCP1a/b double knockout.

Our scrutiny of the transcript profiles and metabolite levels within DCP1a- and DCP1b-knockout cells has unveiled their unique contributions to cellular processes. While DCP1a and DCP1b might exhibit certain common gene targets, the distinct transcriptome profiles suggest their functions are not entirely interchangeable. Pathway analysis indicated that DCP1a and DCP1b may have partially complementary roles in regulating cellular pathways. Additionally, several metabolites showed notable alterations in knockout cells, potentially opening a new avenue for investigating the metabolic consequences of DCP1 knockout. Our findings yield valuable understanding of the distinct functions of DCP1a and DCP1b in human cells and the complex regulatory mechanisms underlying cellular metabolism. The potential targets we identified could be investigated further by quantifying specific transcripts and metabolites, and a larger sample size would facilitate the identification of correlations between differentially expressed genes and differentially abundant metabolites.

Overall, our study provides valuable insights into the molecular mechanisms governing DCP1-mediated decapping in human cells and the coordinated regulation of DCP1a and DCP1b, which has important implications for gene expression and cellular function. Future biochemical characterization of the specific mRNA targets and biological pathways regulated by DCP1a and DCP1b is necessary to fully elucidate their roles. This information may help uncover the precise mechanisms underlying mRNA decapping and gene regulation. Furthermore, studying the dysregulation of DCP1 paralogs, especially in cancer, may reveal potential therapeutic targets and improve treatment strategies.

Data availability

Raw sequencing reads and processed data files containing the read counts and normalized abundance measurements generated in this study were deposited in the GEO under the accession number of GSE230847.

Supplementary data

Supplementary Data are available at *eLife* online.

Acknowledgements

We dedicate this work to Elisa Izaurrealde who passed away during the study period. We thank Sigrun Helms and Heike Budde for their technical assistance. We thank the Metabolomics Core Laboratory, Centers of Genomic and Precision Medicine, National Taiwan University for the instrumental support.

Funding

This work was supported by the Max Planck Society, the Higher Education Sprout Project by the Ministry of Education (MOE) in Taiwan, National Science and Technology Council, Taiwan [MOST109-2311-B-010-001-MY2], and Yen Tjing Ling Medical Foundation [CI-110-17].

Conflict of interest

None declared

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Reviewer #1 (Public review):

Summary & Assessment:

The catalytic core of the eukaryotic decapping complex consists of the decapping enzyme DCP2 and its key activator DCP1. In humans, there are two paralogs of DCP1, DCP1a and DCP1b, that are known to interact with DCP2 and recruit additional cofactors or coactivators to the decapping complex; however, the mechanisms by which DCP1 activates decapping and the specific roles of DCP1a versus DCP1b, remain poorly defined. In this manuscript, the authors used CRISPR/Cas9-generated DCP1a/b knockout cells to begin to unravel some of the differential roles for human DCP1a and DCP1b in mRNA decapping, gene regulation, and cellular metabolism. While this manuscript presents some new and interesting observations on human DCP1 (e.g. human DCP1a/b KO cells are viable and can be used to investigate DCP1 function; only the EVH1 domain, and not its disordered C-terminal region which recruits many decapping cofactors, is apparently required for efficient decapping in cells; DCP1a and b target different subsets of mRNAs for decay and may regulate different aspects of metabolism), there is one key claim about the role of DCP1 in regulating DCP2-mediated decapping that is still incompletely or inconsistently supported by the presented data in this revised version of the manuscript.

Strengths & well-supported claims:

- Through in vivo tethering assays in CRISPR/Cas9-generated DCP1a/b knockout cells, the authors show that DCP1 depletion leads to significant defects in decapping and the accumulation of capped, deadenylated mRNA decay intermediates.
- DCP1 truncation experiments reveal that only the EVH1 domain of DCP1 is necessary to rescue decapping defects in DCP1a/b KO cells.
- RNA and protein immunoprecipitation experiments suggest that DCP1 acts as a scaffold to help recruit multiple decapping cofactors to the decapping complex (e.g. EDC3, DDX6, PATL1, PNRC1, and PNRC2), but that none of these cofactors are essential for DCP2-mediated decapping in cells.
- The authors investigated the differential roles of DCP1a and DCP1b in gene regulation through transcriptomic and metabolomic analysis and found that these DCP1 paralogs target different mRNA transcripts for decapping and have different roles in cellular metabolism and their apparent links to human cancers. (Although I will note that I can't comment on the experimental details and/or rigor of the transcriptomic and metabolomic analyses, as these are outside my expertise.)

Weaknesses & incompletely supported claims:

(1) One of the key mechanistic claims of the paper is that "DCP1a can regulate DCP2's cellular decapping activity by enhancing DCP2's affinity to RNA, in addition to bridging the interactions of DCP2 with other decapping factors. This represents a pivotal molecular mechanism by which DCP1a exerts its regulatory control over the mRNA decapping process." Similar versions of this claim are repeated in the abstract and discussion sections. However,

this claim appears to be at odds with the observations that: (a) in vitro decapping assays with immunoprecipitated DCP2 show that DCP1 knockout does not significantly affect the enzymatic activity of DCP2 (Fig 2C&D; I note that there may be a very small change in DCP2 activity shown in panel D, but this may be due to slightly different amounts of immunoprecipitated DCP2 used in the assay); and (b) the authors show only weak changes in relative RNA levels immunoprecipitated by DCP2 with versus without DCP1 (~2-3 fold change in Fig 3H, where expression of the EVH1 domain, previously shown in this manuscript to fully rescue the DCP1 KO decapping defects in cells, looks to be almost within error of the control in terms of increasing RNA binding). If DCP1 pivotally regulates decapping activity by enhancing RNA binding to DCP2, why is no difference in in vitro decapping activity observed in the absence of DCP1, and very little change observed in the amounts of RNA immunoprecipitated by DCP2 with the addition of the DCP1 EVH1 domain?

In the revised manuscript and in their response to initial reviews, the authors rightly point out that in vivo effects may not always be fully reflected by or recapitulated in in vitro experiments due to the lack of cellular cofactors and simpler environment for the in vitro experiment, as compared to the complex environment in the cell. I fully agree with this of course! And further completely agree with the authors that this highlights the critical importance of in cell experiments to investigate biological functions and mechanisms! However, because the in vitro kinetic and IP/binding data both suggest that the DCP1 EVH1 domain has minimal to no effects on RNA decapping or binding affinity, while the in cell data suggest the EVH1 domain alone is sufficient to rescue large decapping defects in DCP1a/b KO cells (and that all the decapping cofactors tested were dispensable for this), I would argue there is insufficient evidence here to make a claim that (maybe weakly) enhanced RNA binding induced by DCP1 is what is regulating the cellular decapping activity. Maybe there are as-yet-untested cellular cofactors that bind to the EVH1 domain of DCP1 that change either RNA recruitment or the kinetics of RNA decapping in cells; we can't really tell from the presented data so far. Furthermore, even if it is the case that the EVH1 domain modestly enhances RNA binding to DCP2, the authors haven't shown that this effect is what actually regulates the large change in DCP2 activity upon DCP1 KO observed in the cell.

Overall, while I absolutely appreciate that there are many possible reasons for the differences observed in the in vitro versus in cell RNA decapping and binding assays, because this discrepancy between those data exists, it seems difficult to draw any clear conclusions about the actual mechanisms by which DCP1 helps regulate RNA decapping by DCP2. For example, in the cell it could be that DCP1 enhances RNA binding, or recruits unidentified cofactors that themselves enhance RNA binding, or that DCP1 allosterically enhances DCP2-mediated decapping kinetics, or a combination of these, etc; my point is that without in vitro data that clearly support one of those mechanisms and links this mechanism back to cellular DCP2 decapping activity (for example, in cell data that show EVH1 mutants that impair RNA binding fail to rescue DCP1 KO decapping defects), it's difficult to attribute the observed in cell effects of DCP1a/b KO and rescue by the EVH1 domain directly to enhancement of RNA binding (precisely because, as the authors describe, the decapping process and regulation may be very complex in the cell!).

This contradiction between the in vitro and in-cell decapping data undercuts one of the main mechanistic takeaways from the first half of the paper; I still think this conclusion is overstated in the revised manuscript.

Additional minor comment:

- Related to point (1) above, the kinetic analysis presented in Fig 2C shows that the large majority of transcript is mostly decapped at the first 5 minute timepoint; it may be that DCP2-mediated decapping activity is actually different in vitro with or without DCP1, but that this is being missed because the reaction is basically done in less than 5 minutes under the conditions being assayed (i.e. these are basically endpoint assays under these conditions). It

may be that if kinetics were done under conditions to slow down the reaction somewhat (e.g. lower Dcp2 concentration, lower temperatures), so that more of the kinetic behavior is captured, the apparent discrepancy between in vitro and in-cell data would be much less. Indeed, previous studies have shown that in yeast, Dcp1 strongly activates the catalytic step (k_{cat}) of decapping by ~10-fold, and reduces the K_M by only ~2 fold (Floor et al, NSMB 2010). It might be beneficial to use purified proteins here, if possible, to better control reaction conditions.

In their response to initial reviews, the authors comment that they tried to purify human DCP2 from E coli, but were unable to obtain active enzyme in this way. Fair enough! I will only comment that just varying the relative concentration of immunoprecipitated DCP2 would likely be enough to slow down the reaction and see if activity differences are seen in different kinetic regimes, without the need to obtain fully purified / recombinant Dcp2.

<https://doi.org/10.7554/eLife.94811.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Weaknesses & incompletely supported claims:

(1) A central mechanistic claim of the paper is that "DCP1a can regulate DCP2's cellular decapping activity by enhancing DCP2's affinity to RNA, in addition to bridging the interactions of DCP2 with other decapping factors. This represents a pivotal molecular mechanism by which DCP1a exerts its regulatory control over the mRNA decapping process." Similar versions of this claim are repeated in the abstract and discussion sections. However, this appears to be entirely at odds with the observation from in vitro decapping assays with immunoprecipitated DCP2 that showed DCP1 knockout does not significantly affect the enzymatic activity of DCP2 (Figures 2B-D; I note that there may be a very small change in DCP2 activity shown in panel C, but this may be due to slightly different amounts of immunoprecipitated DCP2 used in the assay, as suggested by panel D). If DCP1 pivotally regulates decapping activity by enhancing RNA binding to DCP2, why is no difference in decapping activity observed in the absence of DCP1?

Furthermore, the authors show only weak changes in relative RNA levels immunoprecipitated by DCP2 with versus without DCP1 (~2-3 fold change; consistent with the Valkov 2016 NSMB paper, which shows what looks like only modest changes in RNA binding affinity for yeast Dcp2 +/- Dcp1). Is the argument that only a 2-3 fold change in RNA binding affinity is responsible for the sizable decapping defects and significant accumulation of deadenylated intermediates observed in cells upon Dcp1 depletion? (and if so, why is this the case for in-cell data, but not the immunoprecipitated in vitro data?)

We appreciate the reviewer's thoughtful comments on our paper. The reviewer points out an apparent contradiction between the claim that DCP1a regulates DCP2's cellular decapping activity and the observation that knocking out DCP1a does not significantly affect DCP2's enzymatic activity in vitro. However, it is important to underscore the challenge of reconciling differences between in vitro and in vivo experiments in scientific research. Although in vitro systems provide a controlled environment, they have inherent limitations that often fail to capture the complexities of cellular processes. Our in vitro experiments used immunoprecipitated proteins to ensure the presence of relevant factors, but these

experiments cannot fully replicate the precise stoichiometry and dynamic interactions present in a cellular environment. Furthermore, the limited volume in vitro can actually facilitate reactions that may not occur as readily in the complex and heterogeneous environment of a cell. Therefore, the lack of a significant difference in decapping activity observed in vitro does not necessarily negate the regulatory role of DCP1 in the cellular context. Rather, it underscores our previous oversight of DCP1's importance in the decapping process under in vitro conditions. The conclusions regarding DCP1's regulatory mechanisms remain valid and supported by the presented evidence, especially when considering the inherent differences between in vitro and in vivo experimental conditions. It is precisely because of these differences that we recognized our previous underestimation of DCP1's significance. Therefore, our subsequent experiments focused on elucidating DCP1's regulatory mechanisms in the decapping process

*The authors acknowledge this apparent discrepancy between the in vitro DCP2 decapping assays and in-cell decapping data, writing: "this observation could be attributed to the inherent constraints of in vitro assays, which often fall short of faithfully replicating the complexity of the cellular environment where multiple factors and cofactors are at play. To determine the underlying cause, we postulated that the observed cellular decapping defect in DCP1a/b knockout cells might be attributed to DCP1 functioning as a scaffold." This is fair. They next show that DCP1 acts as a scaffold to recruit multiple factors to DCP2 in cells (EDC3, DDX6, PatL1, and PNRC1 and 2). However, while DCP1 is shown to recruit multiple cofactors to DCP2 (consistent with other studies in the decapping field, and primarily through motifs in the Dcp1 C-terminal tail), the authors ultimately show that *none* of these cofactors are actually essential for DCP2-mediated decapping in cells (Figures 3A-F). More specifically, the authors showed that the EVH1 domain was sufficient to rescue decapping defects in DCP1a/b knockout cells, that PNRC1 and PNRC2 were the only cofactors that interact with the EVH1 domain, and finally that shRNA-mediated PNRC1 or PNRC2 knockdown has no effect on in-cell decapping (Figures 3E and F). Therefore, based on the presented data, while DCP1 certainly does act as a scaffold, it doesn't seem to be the case that the major cellular decapping defect observed in DCP1a/b knockout is due to DCP1's ability to recruit specific cofactors to DCP2.*

The findings that none of the decapping cofactors recruited by DCP1 to DCP2 are essential for decapping in cells further underscore the complexity of the decapping process in vivo. This observation suggests that while DCP1's scaffolding function is crucial for recruiting cofactors, the decapping process likely involves additional layers of regulation that are not fully captured by our current understanding of DCP1. Furthermore, the reviewer mentions that the observed changes in RNA binding affinity (approximately 2-3 fold) in our in vitro experiments seem relatively modest. While these changes may appear insignificant in vitro, their cumulative impact in the dynamic cellular environment could be substantial. Even minor perturbations in RNA binding affinity can trigger cascading effects, leading to significant changes in decapping activity and the accumulation of deadenylated intermediates upon Dcp1 depletion. Cellular processes involve complex networks of interrelated events, and small molecular changes can result in amplified biological outcomes. The subtle molecular variations observed in vitro may translate into significant phenotypic outcomes within the complex cellular environment, underscoring the importance of DCP1a's regulatory role in the cellular decapping process.

So as far as I can tell, the discrepancy between the in vitro (DCP1 not required) and in-cell (DCP1 required) decapping data, remains entirely unresolved. Therefore, I don't think that the conclusions that DCP1 regulates decapping by (a) changing RNA binding affinity (authors show this doesn't matter in vitro, and that the change in RNA binding affinity is very small) or (b) by bridging interactions of cofactors with DCP2 (authors show all tested

cofactors are dispensable for robust in-cell decapping activity), are supported by the evidence presented in the paper (or convincingly supported by previous structural and functional studies of the decapping complex).

We have addressed the reconciliation of differences between in vitro and in vivo experiments in the revised manuscript and emphasized the importance of considering cellular interactions when interpreting our findings.

(2) Related to the RNA binding claims mentioned above, are the differences shown in Figure 3H statistically significant? Why are there no error bars shown for the MBP control? (I understand this was normalized to 1, but presumably, there were 3 biological replicates here that have some spread of values?). The individual data points for each replicate should be displayed for each bar so that readers can better assess the spread of data and the significance of the observed differences. I've listed these points as major because of the key mechanistic claim that DCP1 enhances RNA binding to DCP2 hinges in large part on this data.

Thank you for your feedback. Regarding your comments on the statistical significance of the differences shown in Figure 3H and the absence of error bars for the MBP control, we will address these concerns in the revised manuscript. We'll include individual data points for the three biological replicates and corresponding statistical analysis to more clearly demonstrate the data spread and significance of the observed differences.

(3) Also related to point (1) above, the kinetic analysis presented in Figure 2C shows that the large majority of transcript is mostly decapped at the first 5-minute timepoint; it may be that DCP2-mediated decapping activity is actually different in vitro with or without DCP1, but that this is being missed because the reaction is basically done in less than 5 minutes under the conditions being assayed (i.e. these are basically endpoint assays under these conditions). It may be that if kinetics were done under conditions to slow down the reaction somewhat (e.g. lower Dcp2 concentration, lower temperatures), so that more of the kinetic behavior is captured, the apparent discrepancy between in vitro and in-cell data would be much less. Indeed, previous studies have shown that in yeast, Dcp1 strongly activates the catalytic step (k_{cat}) of decapping by ~10-fold, and reduces the K_M by only ~2 fold (Floor et al, NSMB 2010). It might be beneficial to use purified proteins here (only a Western blot is used in Figure 2D to show the presence of DCP2 and/or DCP1, but do these complexes have other, and different, components immunoprecipitated along with them?), if possible, to better control reaction conditions.

This contradiction between the in vitro and in-cell decapping data undercuts one of the main mechanistic takeaways from the first half of the paper. This needs to be addressed/resolved with further experiments to better define the role of DCP1-mediated activation, or the mechanistic conclusions significantly changed or removed.

We genuinely appreciate the reviewer's insightful comments on the kinetic analysis presented in Figure 2C. Your astute observation regarding the potential influence of reaction duration on the interpretation of in vitro decapping activity, especially in the absence of DCP1, is well-received. The time-sensitive nature of our experiments, as you rightly pointed out, might not fully capture the nuanced kinetic behaviors. In addition, the DCP2 complex purified from cells could not be precisely quantified. In response to your suggestion, we attempted to purify human DCP2 protein from *E. coli*; however, regrettably, the purified protein failed to exhibit any enzymatic activity. This disparity may be attributed to species differences.

Considering the reviewer's valuable insights, our revised manuscript emphasized that purified DCP2 from cells exhibits activity regardless of the presence of DCP1. This adjustment

aims to provide a clearer perspective on our findings and to better align with the nuances of our experimental design and the meticulous consideration of the results.

(4) The second half of the paper compares the transcriptomic and metabolic profiles of DCP1a versus DCP1b knockouts to reveal that these target a different subset of mRNAs for degradation and have different levels of cellular metabolites. This is a great application of the DCP1a/b KO cells developed in this paper and provides new information about DCP1a vs b function in metazoans, which to my knowledge has not really been explored at all. However, the analysis of DCP1 function/expression levels in human cancer seems superficial and inconclusive: for example, the authors conclude that "...these findings indicate that DCP1a and DCP1b likely have distinct and non-redundant roles in the development and progression of cancer", but what is the evidence for this? I see that DCP1a and b levels vary in different cancer cell types, but is there any evidence that these changes are actually linked to cancer development, progression, or tumorigenesis? If not, these broader conclusions should be removed.

Thank you to the reviewer for pointing out that such a description may be misleading. We have removed our previous broader conclusion and revised our sentences. To further explore the potential impact of DCP1a and DCP1b on cancer progression, we examined the association between the expression levels of DCP1a and DCP1b and progression-free interval (PFI). We have incorporated this information into our revised manuscript.

(5) The authors used CRISPR-Cas9 to introduce frameshift mutations that result in premature termination codons in DCP1a/b knockout cells (verified by Sanger sequencing). They then use Western blotting with DCP1a or DCP1b antibodies to confirm the absence of DCP1 in the knockout cell lines. However, the DCP1a antibody used in this study (Sigma D5444) is targeted to the C-terminal end of DCP1a. Can the authors conclusively rule out that the CRISPR/Cas-generated mutations do not result in the production of truncated DCP1a that is just unable to be detected by the C-terminally targeted antibody? While it is likely the introduced premature termination codon in the DCP1a gene results in nonsense-mediated decay of the resulting transcript, this outcome is indeed supported by the knockout results showing large defects in cellular decapping which can be rescued by the addition of the EVH1 domain, it would be better to carefully validate the success of the DCP1a knockout and conclusively show no truncated DCP1a is produced by using N-terminally targeted DCP1a antibodies (as was the case for DCP1b).

Thank you for your insightful comment regarding the validation of our DCP1a/b knockout cell line. We acknowledge your point about the DCP1a C-terminal targeting of the Sigma D5444 antibody used in our Western blot analysis. We agree that we cannot definitively rule out the possibility of truncated DCP1a protein production solely based on the lack of full-length protein detection. To address this limitation, we utilized a commercial information available N-terminally targeted DCP1a antibody (aviva ARP39353_T100) in a Western blot analysis. This will allow us to comprehensively detect any truncated protein fragments remaining after the CRISPR-Cas9-generated frameshift mutation.

Some additional minor comments:

- More information would be helpful on the choice of DCP1 truncation boundaries; why was 1-254 chosen as one of the truncations?*

Thank you for the reviewer's comment and suggestion. Regarding the choice of DCP1 1-254 truncation boundaries based on the predicted structure from AlphaFoldDB (A0A087WT55). We will include this information in the revised manuscript.

• *Figure S2D is a pretty important experiment because it suggests that the observed deadenylated intermediates are in fact still capped; can a positive control be added to these experiments to show that removal of cap results in rapid terminator-mediated degradation?*

Unfortunately, due to our institution's current laboratory safety policies, we are unable to perform experiments involving the use of radioactive isotopes such as ^{32}P . Therefore, while adding the suggested positive control experiment to demonstrate rapid RNA degradation upon decapping would further validate our interpretation, we regret that we cannot carry out this experiment at the moment. However, the observed deadenylated intermediates in Figure S2D match the predicted size of capped RNA fragments, and not the expected sizes of degradation products after decapping. Furthermore, previous literature has well-established that for these types of RNAs, decapping leads directly to rapid 5' to 3' exonuclease-mediated degradation, without producing stable deadenylated intermediates. Thus, we believe that the current data is sufficient to support our conclusion that the deadenylated intermediates retain the 5' cap structure.

Reviewer #2 (Public Review):

Weaknesses:

The direct targets of DCP1a and/or DCP1b were not determined as the analysis was restricted to RNA-seq to assess RNA abundance, which can be a result of direct or indirect regulation by DCP1a/b.

Thank you for raising this important point. In our study, we acknowledge that the use of RNA-seq to assess RNA abundance provides a broad overview of the regulatory impacts of DCP1a and DCP1b. This method captures changes in RNA levels that may arise from both direct and indirect regulatory actions of these proteins. While we did not directly determine the targets of DCP1a and DCP1b, the data obtained from our RNA-seq analysis serve as a foundational step for future targeted experiments, which could include techniques such as RIP-seq, to delineate the direct targets of DCP1a and DCP1b more precisely. We believe that our current findings contribute valuable information to the field and pave the way for these subsequent analyses.

P-bodies appear to be larger in human cells lacking DCP1a and DCP1b but a lack of image quantification prevents this conclusion from being drawn.

Thank you for the reviewer's valuable feedback. We have addressed the reviewer's concern regarding P-bodies' size in human cells lacking DCP1a and DCP1b. We have now performed image quantification and can confirm that P-bodies are indeed larger in these cells.

The lack of details in the methodology and figure legends limit reader understanding.

We acknowledge the reviewer's concerns regarding the level of detail provided in the methodology and figure legends. To address this, we are committed to enhancing both sections with additional details and clarifications in our revised manuscript. Thank you for bringing this to our attention.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) To me, the second half of the paper comparing DCP1a and DCP1b is in many ways distinct from the first half and could stand on its own as an interesting paper if this comparative analysis is explored a little deeper (maybe by validating some of the differences in decay observed for individual mRNAs targeted by DCP1a versus DCP1b, by measuring and comparing the decay rates of some individual transcripts under differential control by DCP1a vs b?), and revising the conclusions about links to cancer as mentioned above. I think these later comparative results in the paper present the most new and interesting data concerning DCP1 function in humans (especially since I think the mechanistic conclusions from the first half aren't well supported yet or are at least inconsistent), but when I read these later sections of the paper I struggle to understand the key takeaways from the transcriptomic and metabolomic data.

Thank you for the reviewer's suggestions. Estimating the decay rates of individual transcripts within the transcriptomes of DCP1a_KO, DCP1b_KO, and wild type can provide insight into the direct targets of DCP1a or DCP1b. However, this requires either time-series RNA-seq or specialized sequencing technologies such as Precision Run-On sequencing (PRO-seq) or RNA Approach to Equilibrium Sequencing (RATE-Seq). Unfortunately, we lack the necessary dataset in our project to estimate the decay rates for the potential targets identified in our RNA-seq data. Despite this limitation, we acknowledge the potential of this approach in identifying the true targets of DCP1a and DCP1b and have included this idea in our discussion.

(2) I think it would be helpful to add a little more descriptive or narrative language to the figure legends (I know some of them are already quite long!) so that readers can follow the general idea of the experiment through the figure legend as well as the main text; as written, the figure legends are mostly exclusively technical details, so it can be hard to parse what experiment is being carried out in some cases.

Thank you for the reviewer's suggestion, we will strive to improve the language of the figure legends to include technical details while clearly conveying the main idea of the experiment. We will ensure that the language of the figure legends is more readable and comprehensible so that readers can more easily parse what experiment is being carried out.

Reviewer #2 (Recommendations For The Authors):

Suggestions for improved or additional experiments, data, or analyses:

The use of RNA-seq to measure RNA abundance in DCP1a and/or b knockout cells can give some insight into both the indirect and direct effects of DCP1a/b on gene expression but cannot identify the direct targets of these genes. Rather, global analysis of RNA stability or capturing uncapped RNA decay intermediates would allow the authors to conclude they have identified direct targets of DCP1a and/or b. Without such analyses, the interpretation of these data should be scaled back to clearly state that RNA levels can be altered through indirect effects of DCP1a/b absence throughout the text.

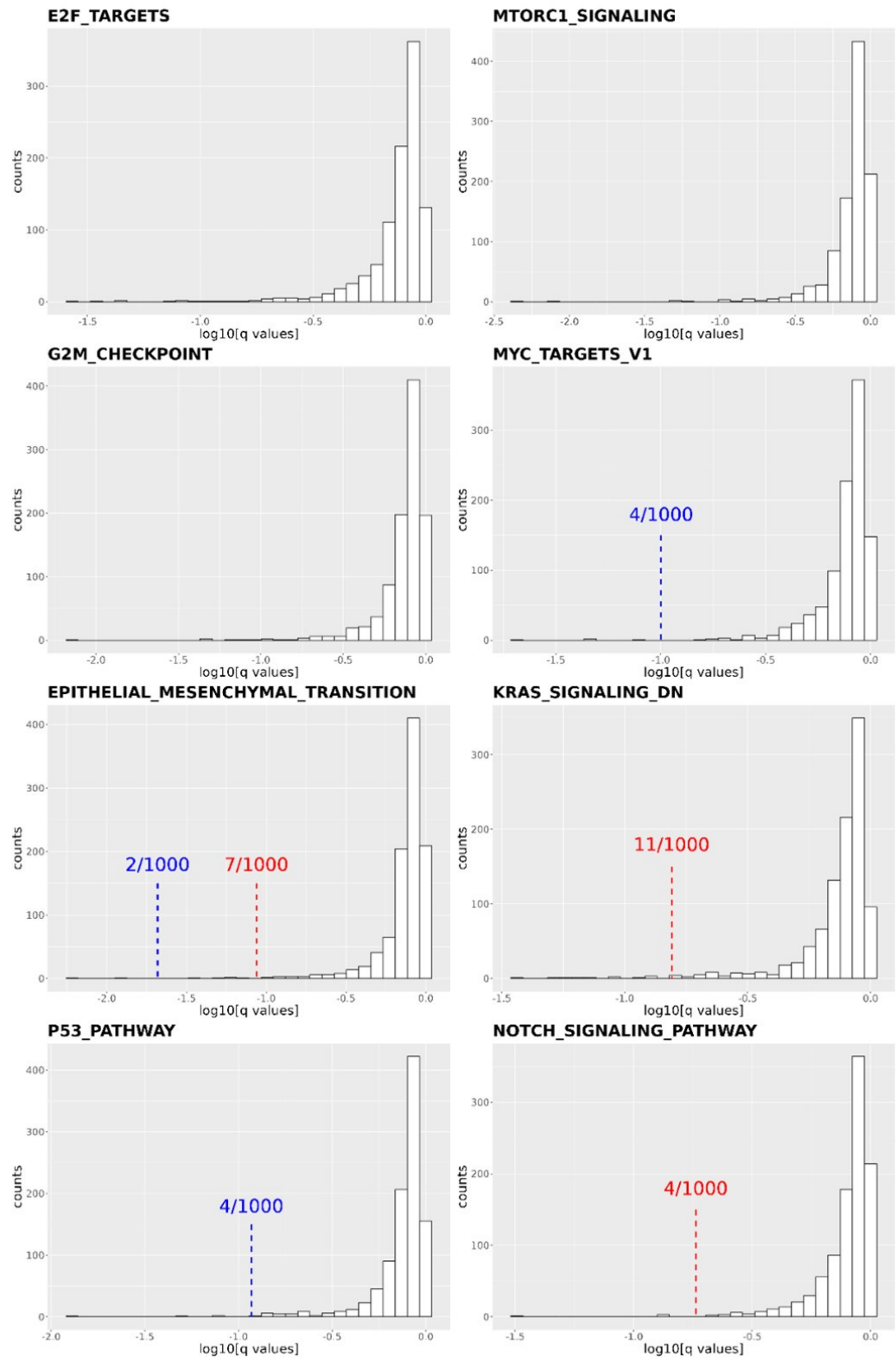
We appreciate the reviewer's suggestion. We have modified our sentences to emphasize that the dysregulated genes could be caused by both direct and indirect effects.

A control/randomly generated gene list should be analyzed for GO terms to determine whether the enrichment of cancer-related pathways in the differentially expressed genes in the DCP1a/b knockout cells is meaningful.

Thank you for the reviewer's comment. We shuffled our gene list and reperformed the pathway enrichment analysis in Figure 4C and 4D 1,000 times. We focused on the following cancer-related pathways: E2F targets, MTORC1 signaling, G2M checkpoint, MYC target V1, EMT transition, KRAS signaling DN, P53 pathway, and NOTCH signaling pathways. We then calculated how many times the q-values obtained from the shuffled gene list were more significant than the q-value obtained from our real data. In four of the eight pathways (E2F targets, MTORC1 signaling, G2M checkpoint, and MYC target v1), none of the shuffled gene lists resulted in a q-value smaller than the real one. In the other four pathways (EMT transition, KRAS signaling DN, P53 pathway, and NOTCH signaling pathways), the q-values were smaller than the real q-value 2, 11, 4, and 4 times out of the 1000 shuffles. Based on the shuffled results, we conclude that the transcriptome of DCP1a/b knockout cells is statistically enriched in these cancer-related pathways.

Author response image 1.

Distribution of q-values resulting from the Gene Set Enrichment Analysis (GSEA) conducted on 1,000 shuffled gene lists for eight cancer-related pathways. The q-values derived from Figure 4C and 4D are indicated by red (DCP1a_KO) and blue (DCP1b_KO) dashed lines, respectively. Some q-values derived from Figure 4C are too small to be labeled on the plots, such as in E2F targets (q value: $5.87\text{E-}07$), MTORC1 signaling (q values: $6.59\text{E-}07$ and $1.58\text{E-}06$ for DCP1a_KO and DCP1b_KO, respectively), MYC target V1 (q value: 0.004644174 for DCP1a_KO), etc. The numbers x/1000 indicate how often the shuffled q-values were smaller than the real q-value out of 1,000 permutations.



Comparisons of the DCP1a and/or b knockout RNA-seq results should be done to published datasets such as those published by Luo et al., Cell Chemical Biology (2021) to determine whether there are common targets with DCP2 and validate the reported findings.

Thank you for reviewer's suggestion. We compared the upregulated genes from DCP1a_KO, DCP1b_KO, and DCP1a/b_KO cell lines with the 91 targets of DCP2 identified by Luo et al. in Cell Chemical Biology (2021). Only EPPK1 was found to be overlapped between the potential DCP1b_KO targets and the targets of DCP2. No genes were found to be overlapped between the potential DCP1a_KO targets and the targets of DCP2. However, three genes, TES, PAX6, and C18orf21, were found to be overlapped between the significantly upregulated DEGs of DCP1a/b_KO and the targets of DCP2. We have included this information in the discussion section.

The RNA tethering assays are not clear and are difficult to interpret without further controls to delineate the polyadenylated and deadenylated species.

Thank you for the reviewer's feedback. We acknowledge that the reviewer might harbor some doubts regarding the outcomes of the RNA tethering assays. Nonetheless, this methodology is well-established and has also found extensive application across many studies. We are committed to enhancing the clarity of our experiment's details and results within the figure legends and textual descriptions.

The representative images of p-bodies clearly show that DCP1a/b KO cells have larger p-bodies than the wild-type cells. The authors should quantify p-body size in each image set as the current interpretation of the data is that there is no difference in size or number of p-bodies, but the data suggest otherwise.

Thank you very much for the reviewer's insightful comments and for drawing our attention to the need to quantify p-body sizes in DCP1a/b KO and wild-type cells. We agree with the reviewer's assessment that the representative images suggest a difference in p-body size between DCP1a/b KO cells and wild-type cells, which we initially overlooked. We will revise our manuscript accordingly to include these findings, ensuring that our interpretation of the data aligns with the observed differences.

Statistical analysis of the Figure 2C results should be included because the difference between the wild-type and Dcp1a/b KO cells with GFP-DCP2 looks significantly different but is interpreted in the text as not significant.

Thank you for pointing out the need for a statistical analysis of the results shown in Figure 2C. We acknowledge that the visual difference between the wild-type and Dcp1a/b KO cells with GFP-DCP2 suggests a significant variation, which may not have been clearly communicated in our text. We will conduct the necessary statistical analysis to substantiate the observations made in Figure 2C. Furthermore, we would like to emphasize that our primary focus was to demonstrate that purified DCP2 within cells retains its activity even in the absence of DCP1. This critical point will be highlighted and clarified in the revised version of our manuscript to prevent any misunderstanding.

Recommendations for improving the writing and presentation:

Additional context including what is known about the role of dcp1 in decapping from the decades of work in yeast and other model organisms should be incorporated into the introduction and discussion sections.

Thank you for the reviewer's suggestion. We will incorporate additional context about the function and significance of DCP1 in decapping processes within our revised manuscript's introduction and discussion sections.

Details should be provided within the figure legends and methods section on experimental approaches and the number of replicates and statistical analyses used throughout the manuscript. For example, it is not clear whether western blots or RNA-IP experiments were performed more than once as representative images are shown.

Thank you for the reviewer's suggestion. In the figure legends and methods section, we will provide more details about the experimental methods, number of replicates, and statistical analyses. Regarding the Western blots and RNA-IP experiments the reviewer mentioned, we performed multiple experiments and presented representative images in the manuscript. We will clarify this in the revised manuscript to eliminate potential confusion.

The rationale for performing metabolic profiling is not clear.

We appreciate the reviewer's thoughtful feedback. The rationale behind conducting metabolic profiling in our study is rooted in its efficacy as a valuable tool for deciphering the consequences of specific gene mutations, particularly those closely associated with phenotypic changes or final metabolic pathways. Our objective is to utilize metabolic profiling to unravel the distinct biofunctions of DCP1a and DCP1b. By employing this approach, we aim to gain insights into the intricate metabolic alterations that result from the absence of these genes, thereby enhancing our understanding of their roles in cellular processes. We recognize the necessity of clearly presenting this rationale and promise to bolster the articulation of these points in the revised version of our manuscript to ensure the clarity and transparency of our research motivation.

Details in the methods section should be included for the CRISPR/Cas9-mediated gene editing validation. The Sanger sequencing results presented in Figure S1b should be explained. The entire western blot(s) should be shown in Figure S1A to give confidence the Dcp1a/b KO cells are not expressing truncated proteins and the epitopes of the antibodies used to detect Dcp1a/b should be described. The northern blot probes should be described and sequences included. The transcriptomics method should be detailed.

Thank you for your feedback, in the revised manuscript we will detail the CRISPR/Cas9 gene editing validation, explain the Sanger sequencing results in Figure S1b, show the full Western blot in Figure S1A to confirm that the Dcp1a/b knockout cells are not expressing truncated proteins, describe the Northern blot probes used, and detail the transcriptomics method, all to ensure clarity and comprehensiveness in our experimental procedures and results.

A diagram showing the RNA tethering assays with labels corresponding to all blots/gels should be provided.

Thank you for your suggestion. We will provide a diagram showing the RNA tethering assays with labels corresponding to all blots/gels in our revised manuscript. This will help readers better understand our experimental design and results.

The statement, "This suggests that the disruption of the decapping process in DCP1a/b-knockout cells results in the accumulation of unprocessed mRNA intermediates" regarding the results of the RNA-seq assay is not supported by the evidence as RNA-seq does not measure RNA decay intermediates or RNA decay rates.

Thank you for the reviewer's comment. We agree with that RNA-seq experiments indeed do not directly measure RNA decay intermediates or RNA decay rates. Our statement could have caused confusion, and we have therefore removed this sentence from the manuscript.

Minor corrections to the text and figures:

Figure S6A is uninterpretable as presented.

Thank you for the reviewer's valuable feedback. We have taken note and made improvements. We have simplified Figure S6A to enhance its interpretability, hoping that the current version will make it easier for the readers to understand.

<https://doi.org/10.7554/eLife.94811.2.sa0>